MOM Assembly Manual

Movable Objective Microscope®
(MOM®)

Assembly Instruction Manual (June 2021)

SUTTER INSTRUMENT
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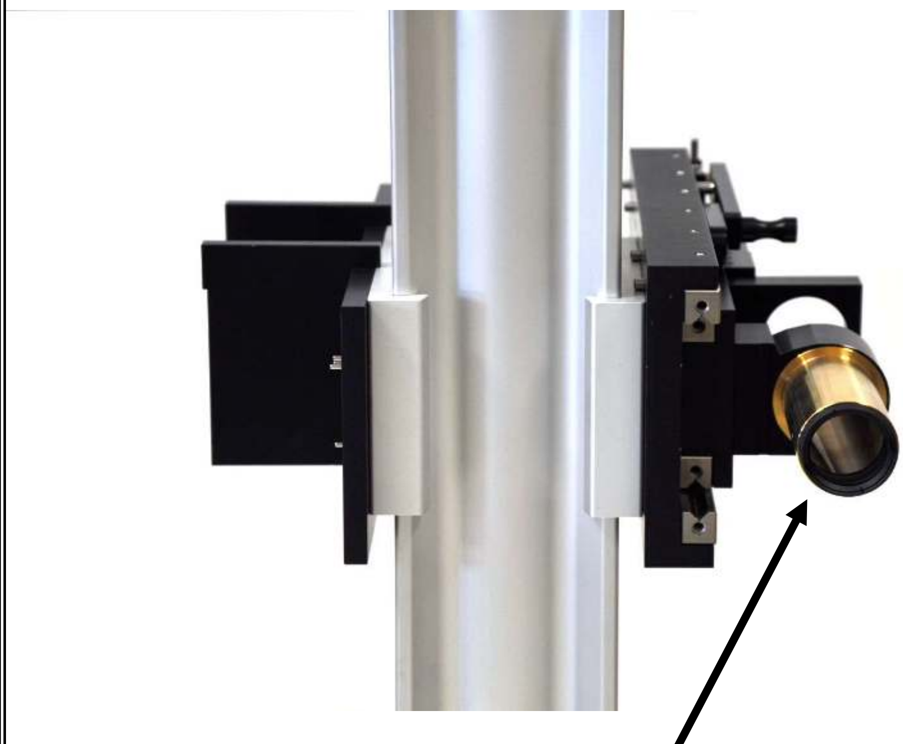
Attach the X-95 rail to the Sutter base using four M6x20mm screws in the indicated locations (arrows).

Slider including the clamp that attaches to the X-95 rail.

Olympus vertical illuminator mount including the clamp that attaches it to the X-95 rail.

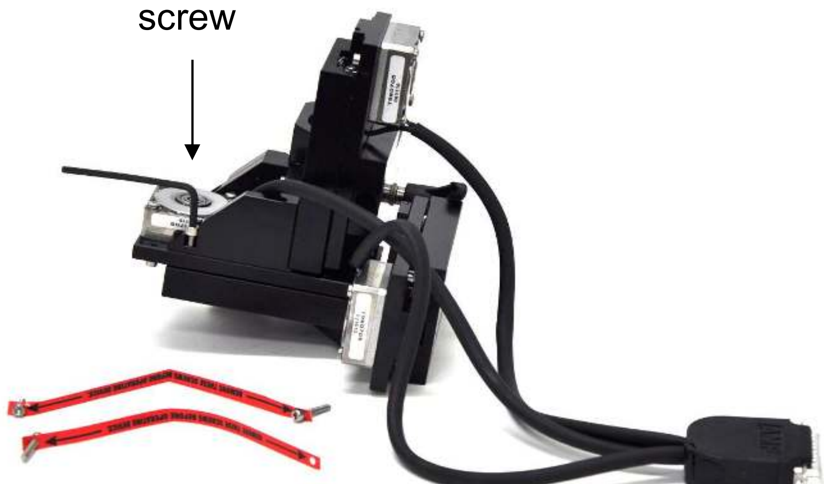
Clamp the Slider and Olympus mount on to the X-95 rail at the same height.



The main microscope body mounts and rotates on the bronze stator.

Next remove the main body of the microscope from its packaging. Remove the three ship screws and install one M4x10mm screw to complete assembly.

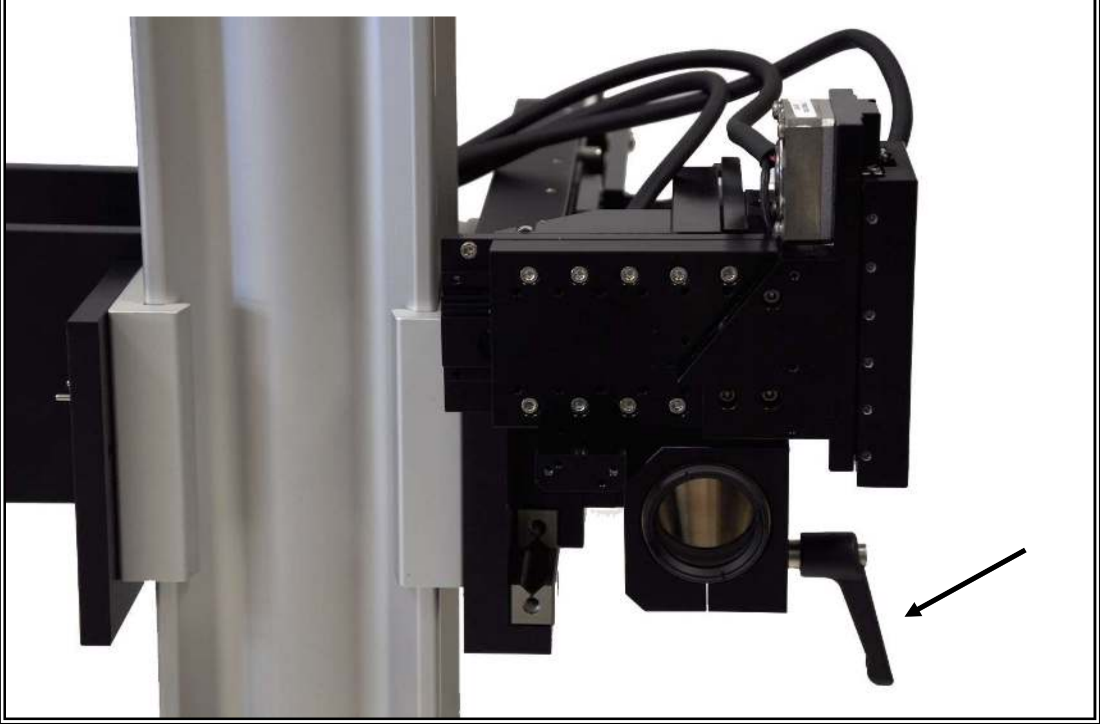
Addition of M4 screw



Removed ship screws

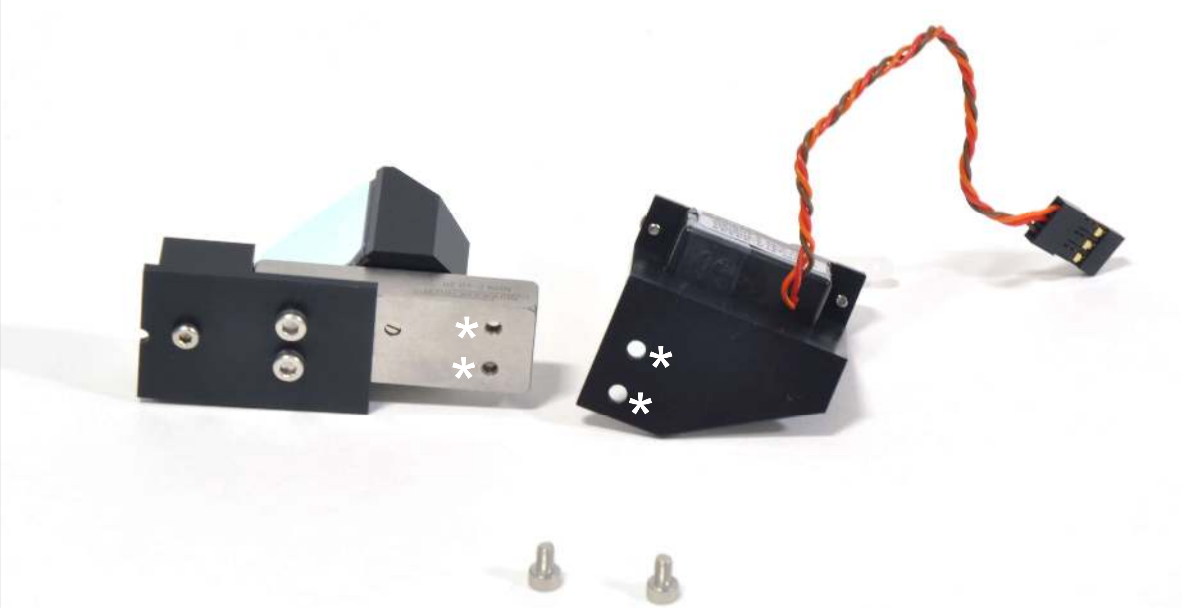
The main microscope body is shown ready to be attached.
The rotator (arrow) slides over the bronze stator.

Tighten the clamp screw (arrow) to fix the rotator on the stator.

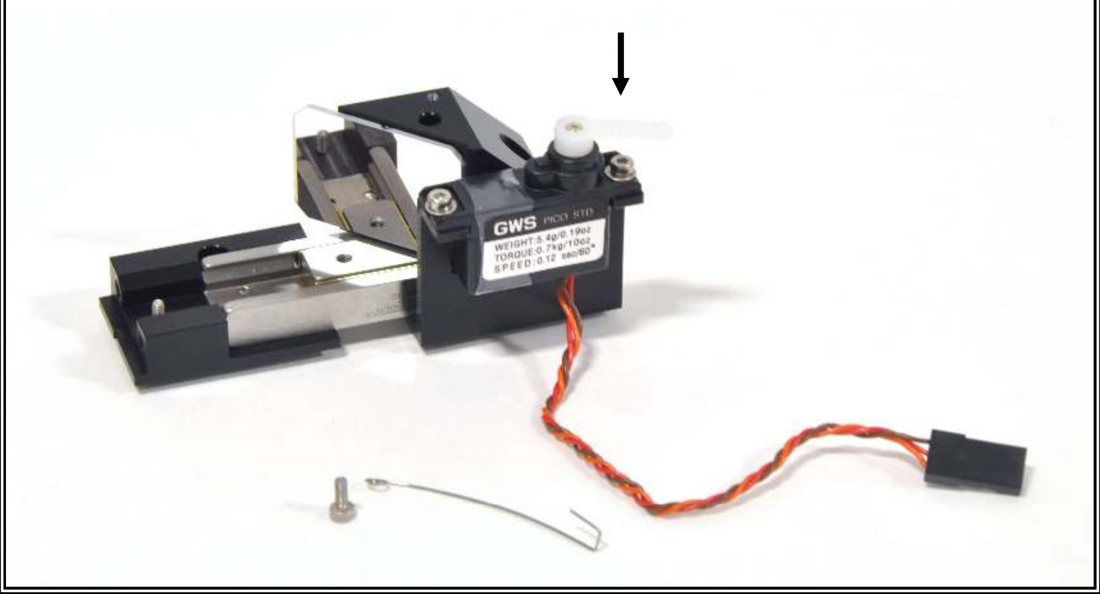


The dichroic slider, servo and associated hardware are shown.

Attach the slider holding the dichroic to the mounting plate attached to the servo with two M2.5x4mm screws (*).



Attach the arm on the servo to the dichroic slider with a spring (shown below). The hook on the spring goes through the servo arm (arrow). The other end of the spring attaches to the dichroic slider with an M2x5mm screw (*).



The completed dichroic slider with hardware for attaching it to the main body of the microscope is shown.
Place this aside for later.

For Short-Path Detector mounting instructions, please see Appendix A.

For Wide-Path (Janelia) Detector mounting instructions, please see Appendix B.

The objective and detector path mirror holder and mounting screws are shown below.

Care must be taken to avoid touching the laser blocking filter shipped in the opening for the detector path (white arrow).

Attach the objective/detector mirror holder in the oultined area with two M3x8mm screws (*).

The objective/detector mirror holder is shown correctly attached.

The objective/detector mirror holder is shown correctly attached from a front view.

The objective is threaded into this ring before installing in the microscope. The ring shown fits Nikon and Leica objectives and the Olympus Super 20x water objective. An alternative ring is available that fits conventional Zeiss and other Olympus objectives with RMS threads (smaller diameter hole).

Mounting objectives.

The objective mount allows the user to quickly change between different objectives.

The objective (not shown) is first threaded onto a brass holder which can be inserted into the objective holder mount.

The objective holder is then rotated clockwise into the mount. The user should not adjust the screws indicated by the arrows to adjust clamping force.

The YZ mirror is shown glued to the YZ mirror holder with the mounting hardware in a bag.

Attach the YZ mirror with one M3x10mm screw (*) and two M2x10mm alignment pins (arrows) in the outlined area.

The YZ mirror is shown correctly attached.

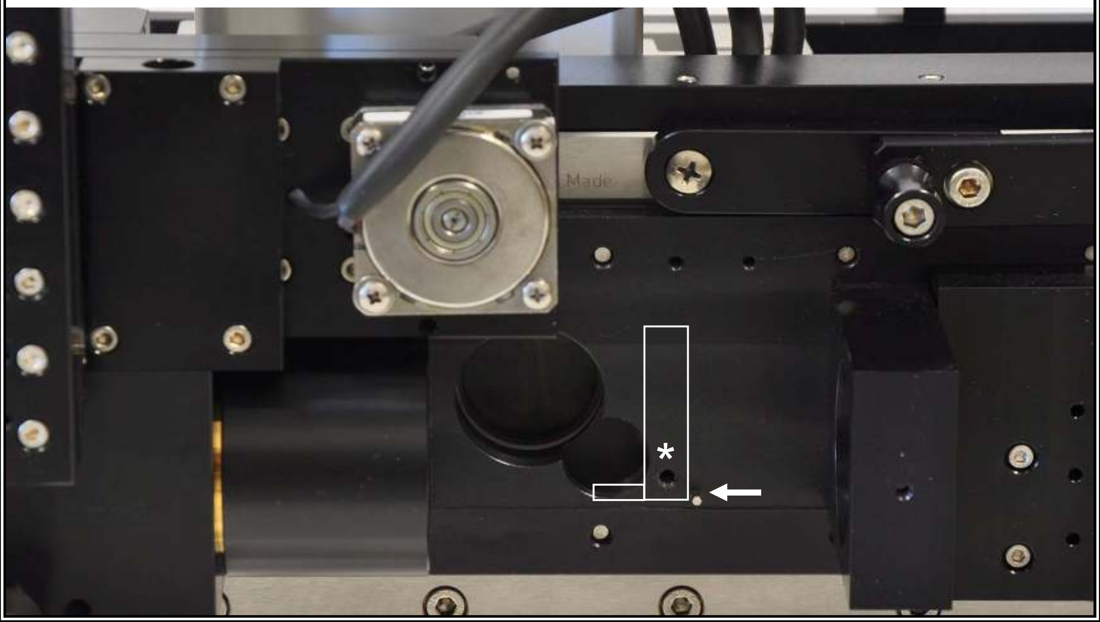
The Y mirror is shown glued to the Y mirror holder with the mounting hardware in a bag.

Attach the Y mirror holder with two M3x10mm screws (*) and two M2x10mm alignment pins (arrows) in the outlined area.

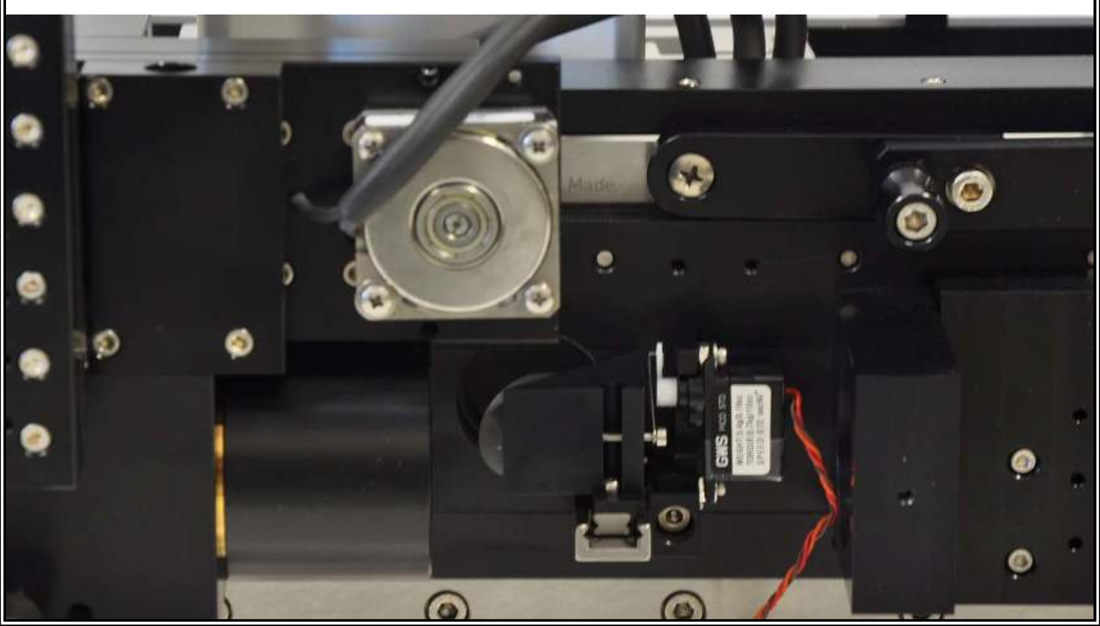
The Y mirror holder is shown correctly attached.

The CCD path mirror, slider and mount with hardware in a bag are shown.

Attach the CCD path mirror and servo to the outlined area with one M4x25mm screw (*) on the side of the Slider. Justify to the bottom ridge on the slider and to an M2x8mm alignment pin at the rear (arrow).



The CCD path mirror and servo are shown correctly attached and in position for two-photon imaging.

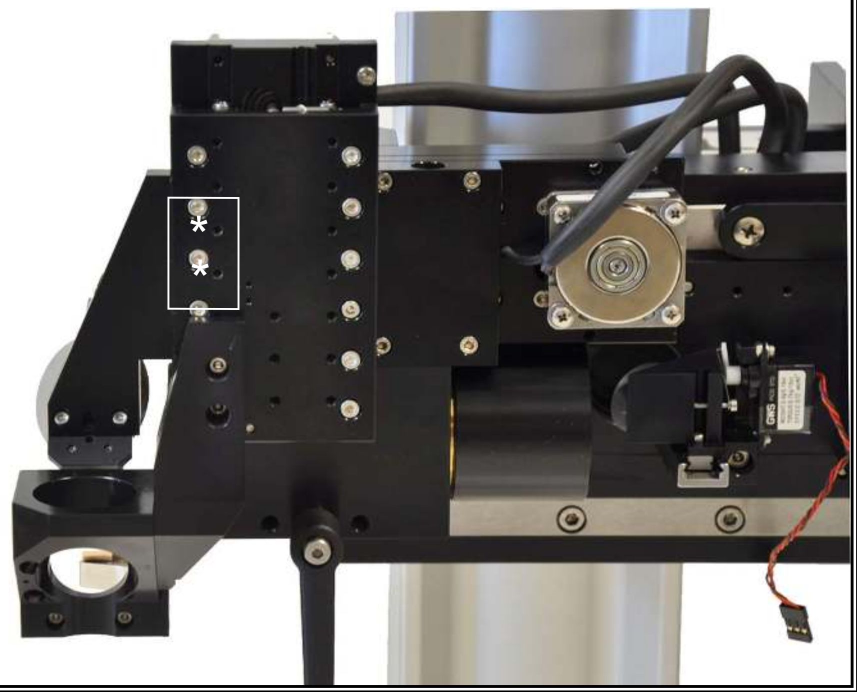


A 40mm diameter, f140mm achromatic lens is shipped separately and needs to be installed in the detector path.

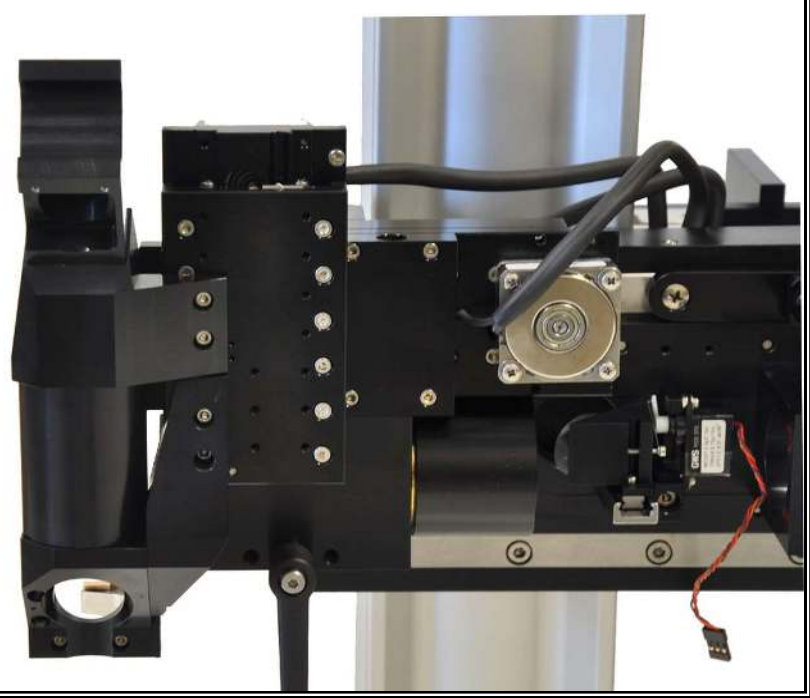
It simply sits on top of the light shield tube, with the convex side of the lens down.

The upper two pieces of the detector path slide down over the lens and light shield to form the complete path. The light shield and lens are kept in place by pressure from the parts below.

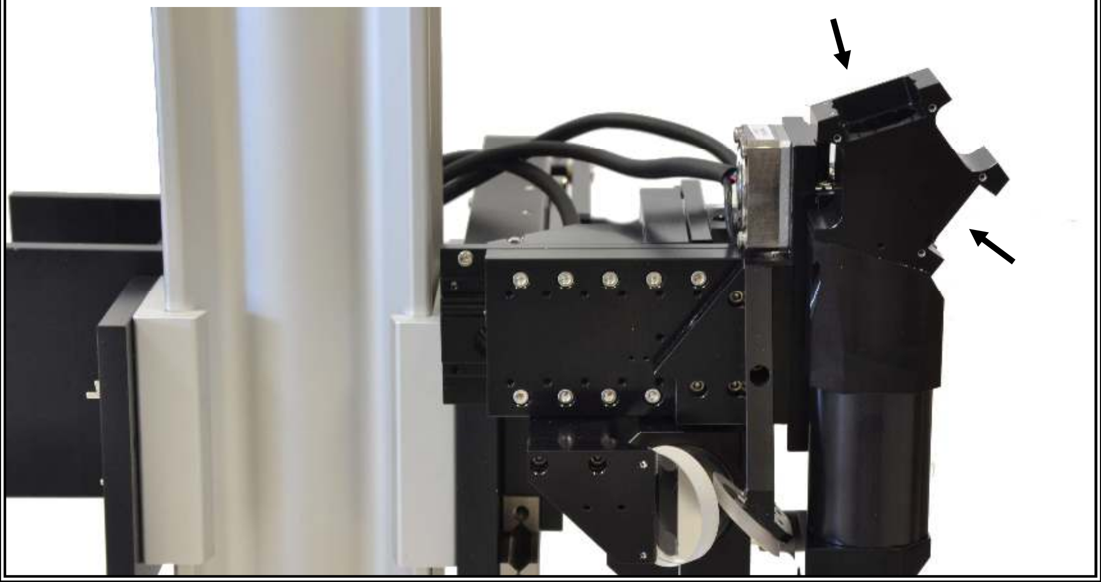
Mount the detector path in the outlined area with two M3x8mm screws (*). Justify the mount to the corner of the Z axis moving slide.



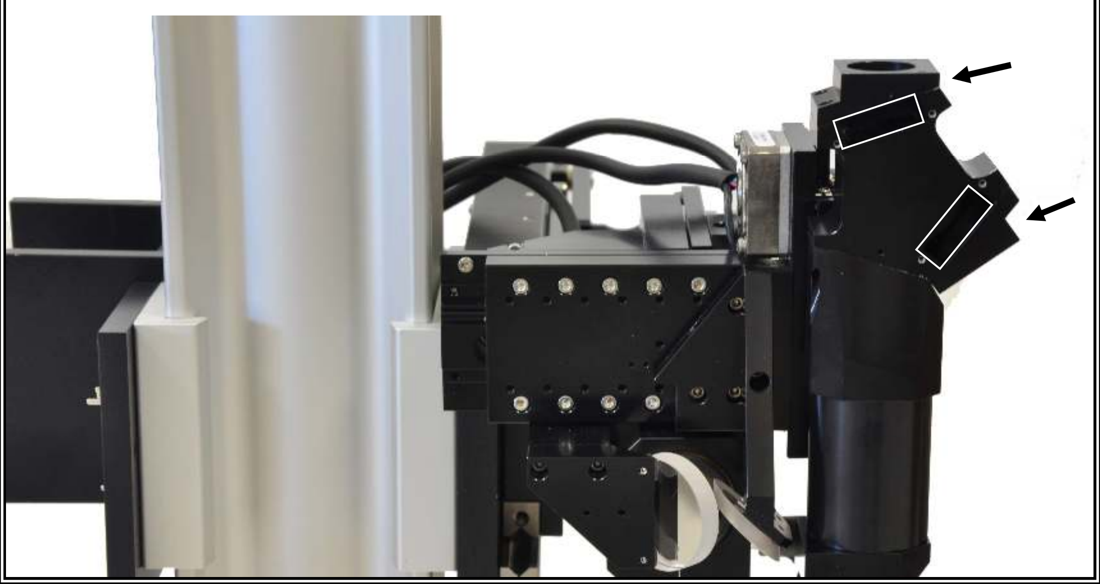
The correctly mounted detector path is shown.



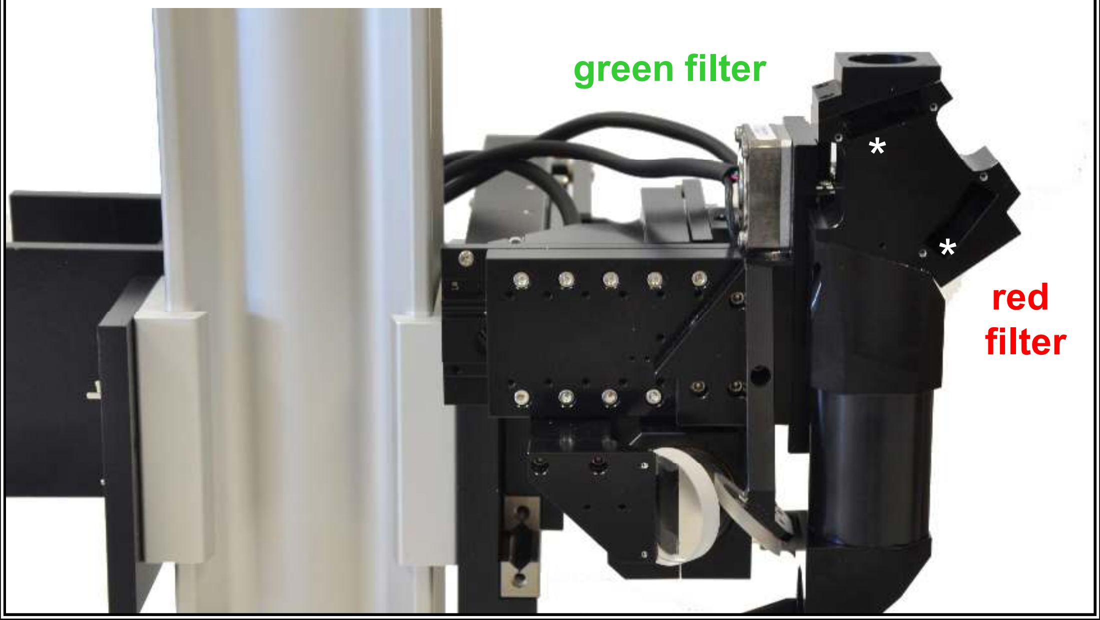
View of the detector head correctly installed from the front side of the scope. Aspherical lens, emission filter and photomultiplier tube (PMT) mounting occurs at the indicated (arrows) locations.



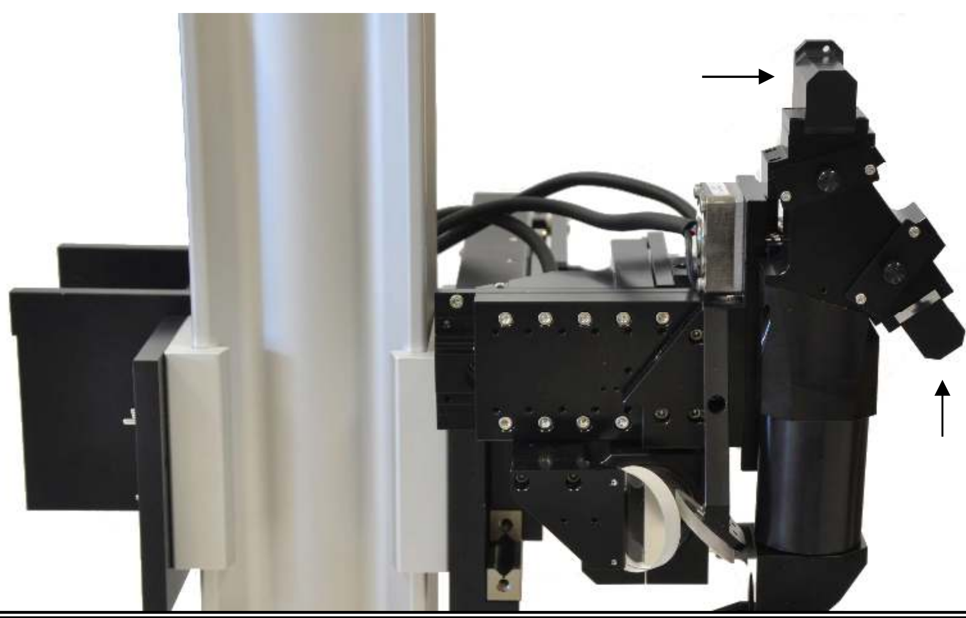
View of the detector head with aspherical lens holders installed (arrows). Emission filters are located in the outlined areas.



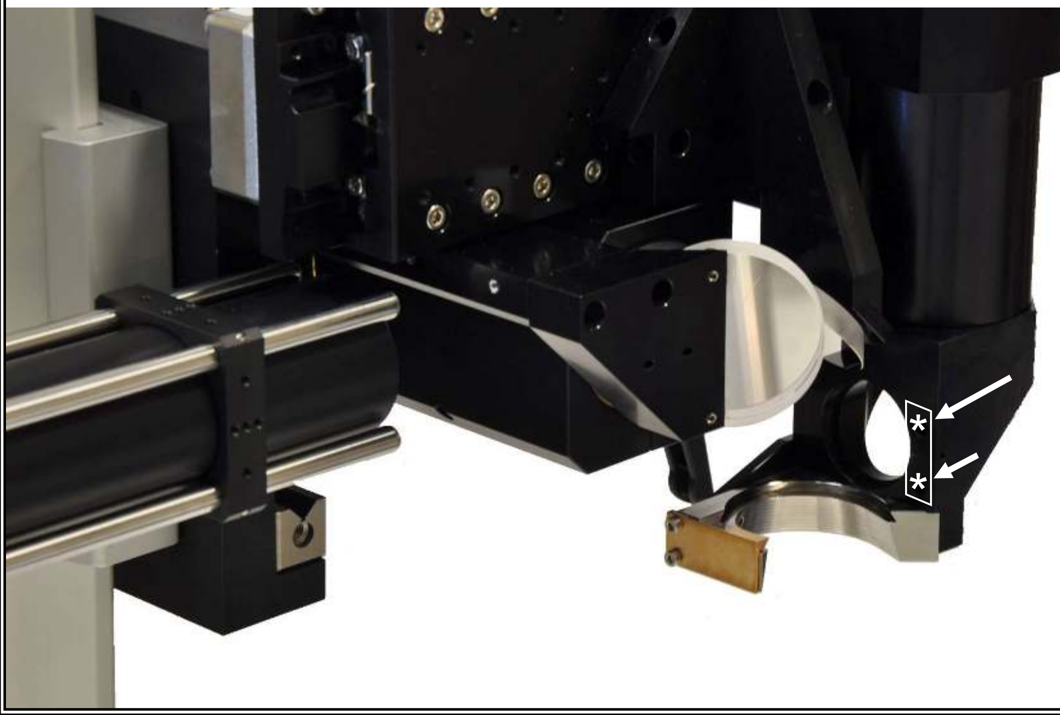
The detector head with emission filter retainers installed (*) is shown.



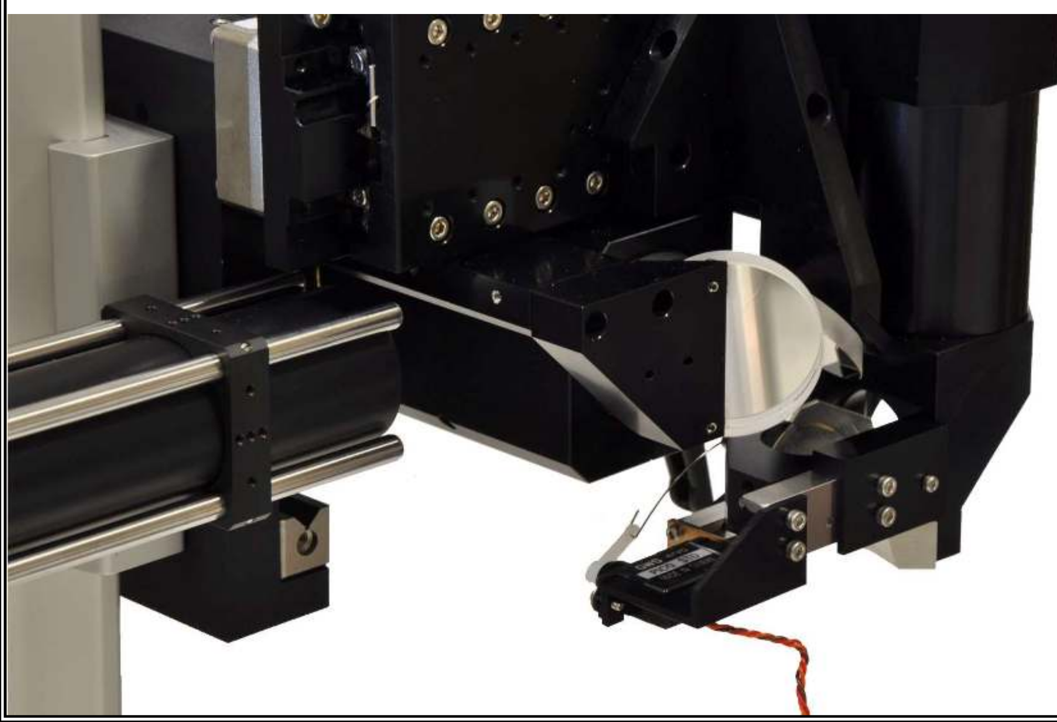
The detector head with side-on PMT holders installed (arrows). All detector head components are mounted with M2X5 screws. This is the standard shipping configuration for the detector head.



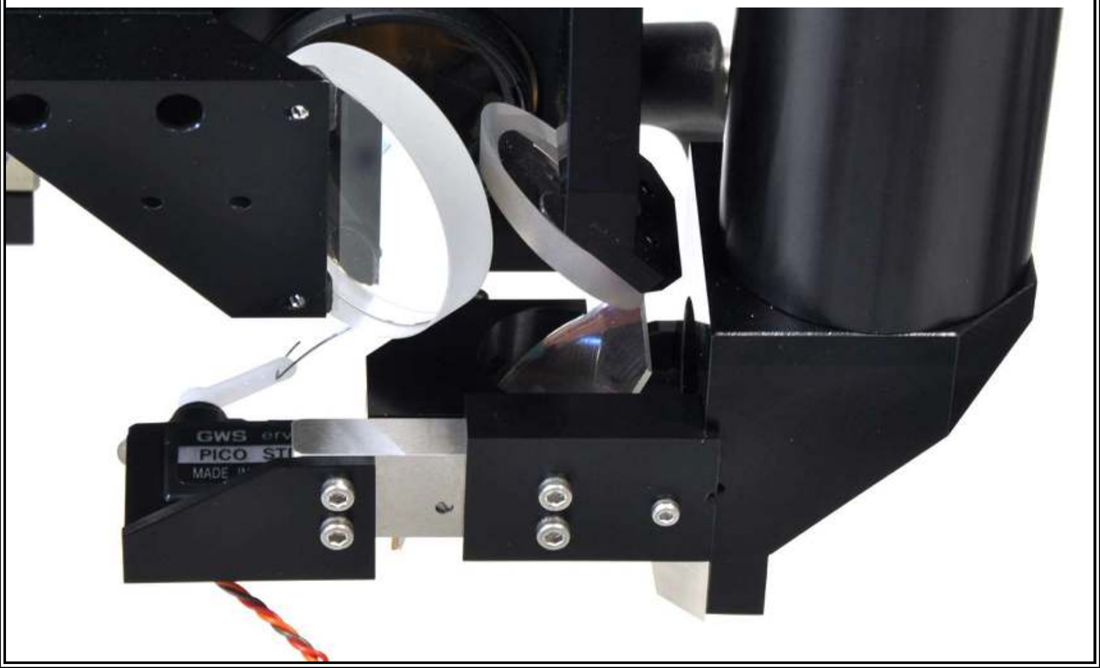
Attach the sliding dichroic mount to the outlined area with two M3x8mm screws (*). The through holes are indicated with white arrows.



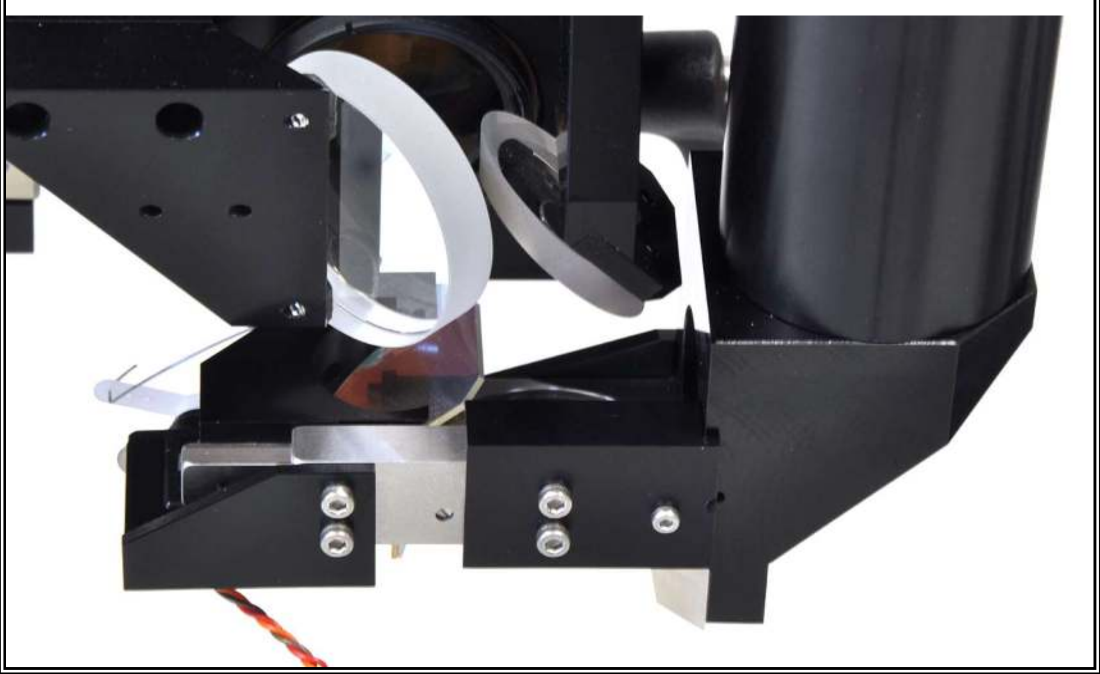
The correctly attached sliding dichroic mount is shown.



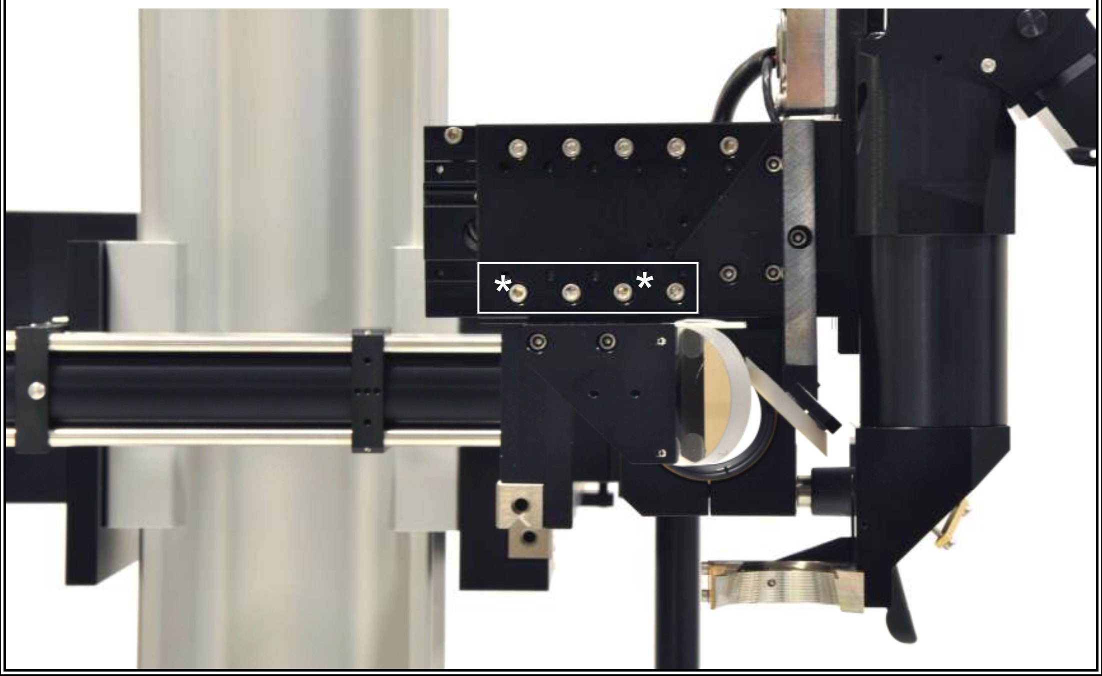
The dichroic slider is shown in place and positioned for PMT use.



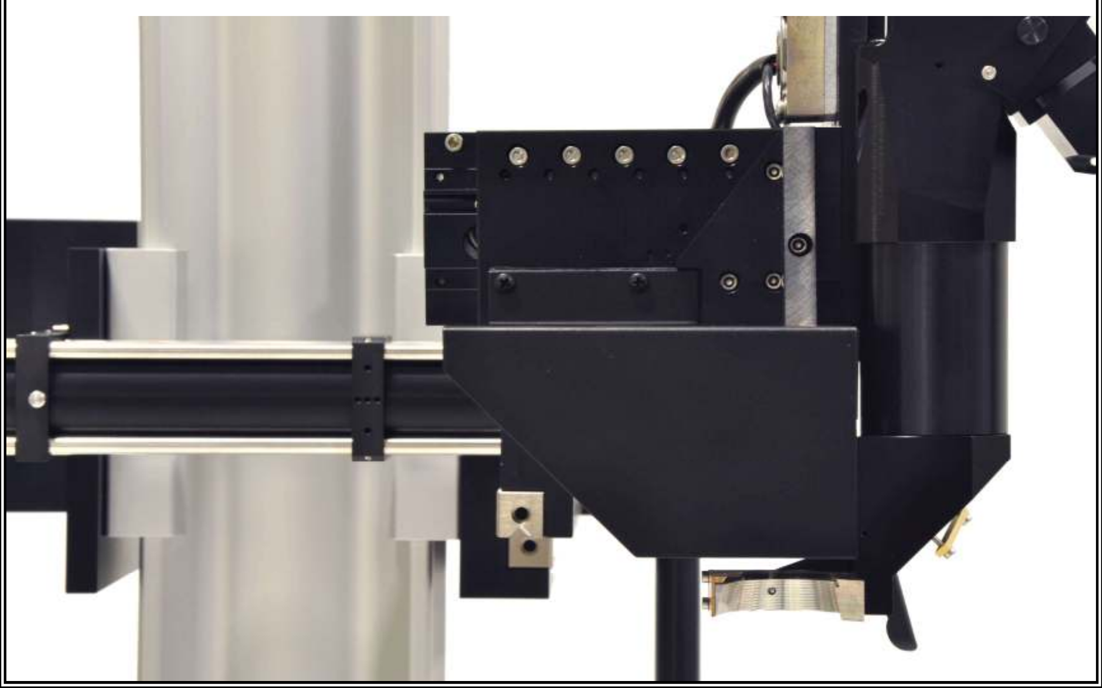
The dichroic slider is shown in place and positioned for standard fluorescence use.



The front cover is attached in the oultined location with two M3x8mm black phillips head screws (*).

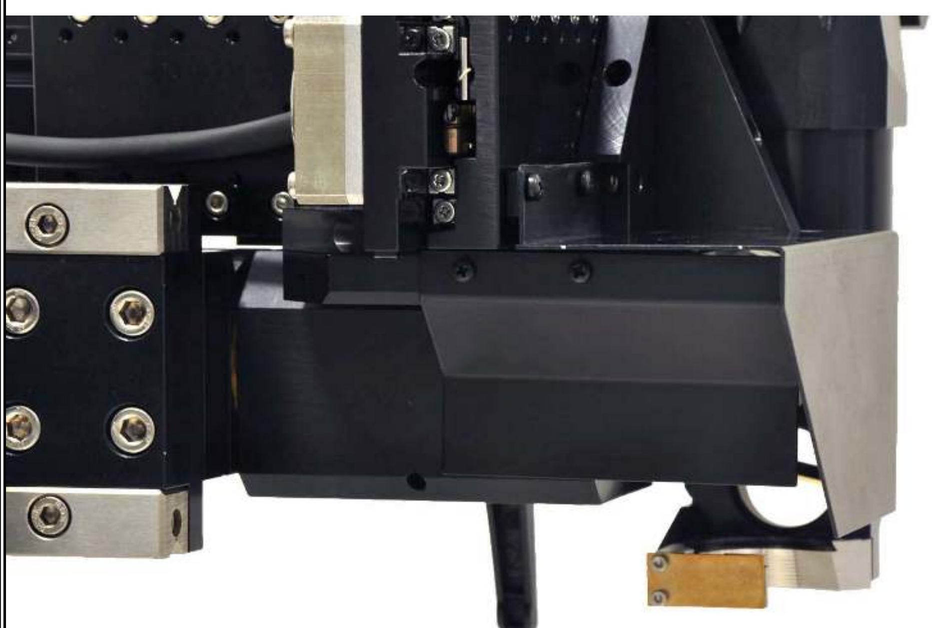


Correct installation of the front cover is shown.

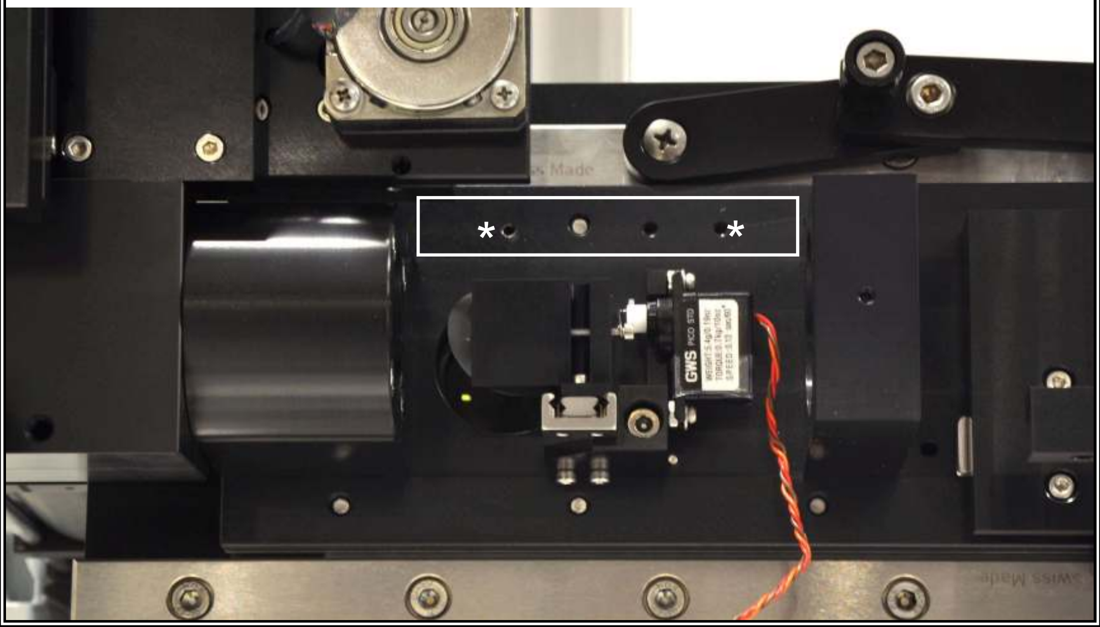


The side cover is attached in the oultined location with two M2.5x6mm black phillips head screws (*).

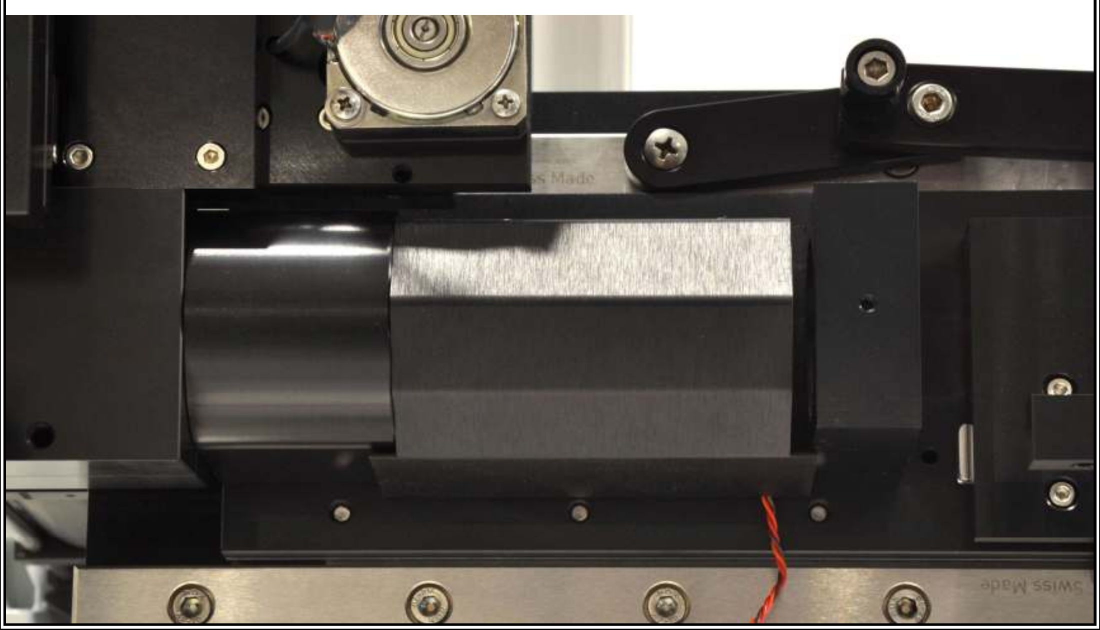
Correct installation of the side cover is shown.



The CCD mirror cover is attached in the oultined location with two M3x8mm black phillips head screws (*).

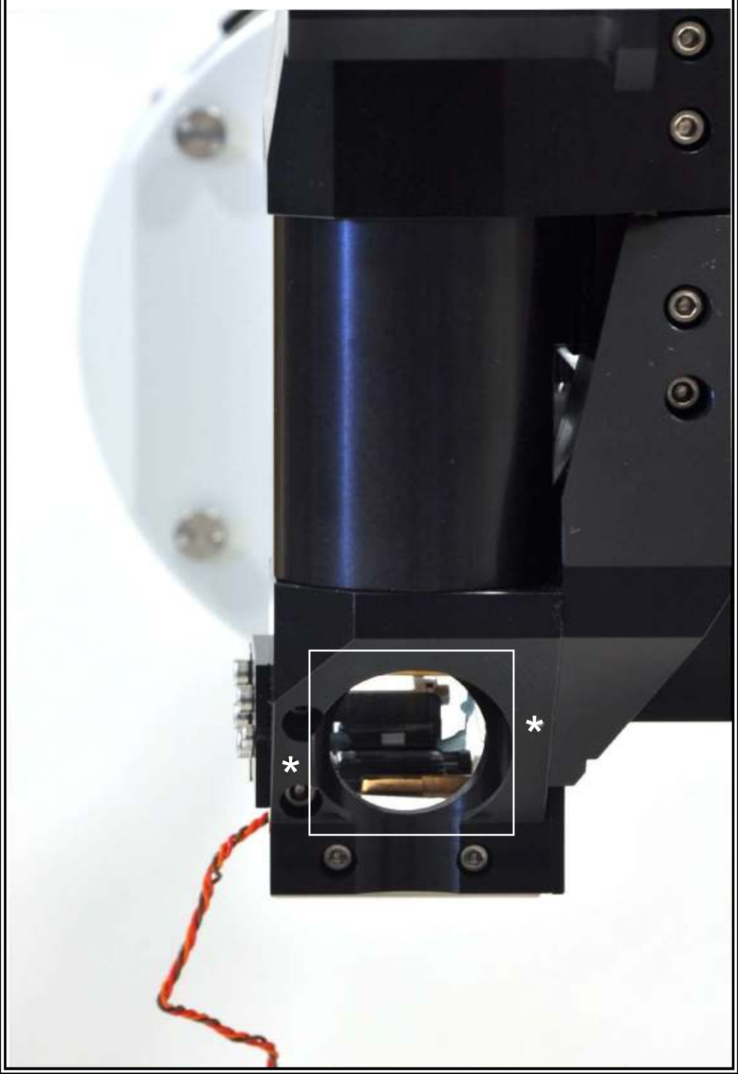


Correct installation of the CCD mirror cover is shown.

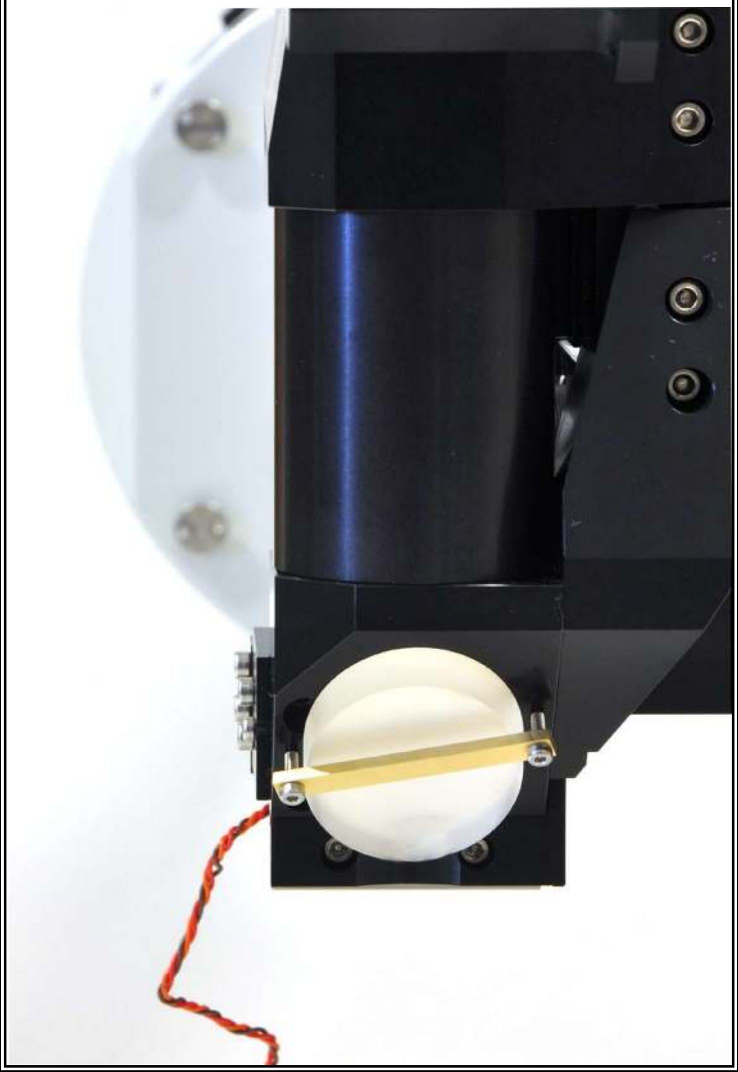


The 45-degree, eliptical, detector-path mirror is shown below.

Attach the detector-path mirror in the outlined area with two M2x10 screws (*) and the clamp.

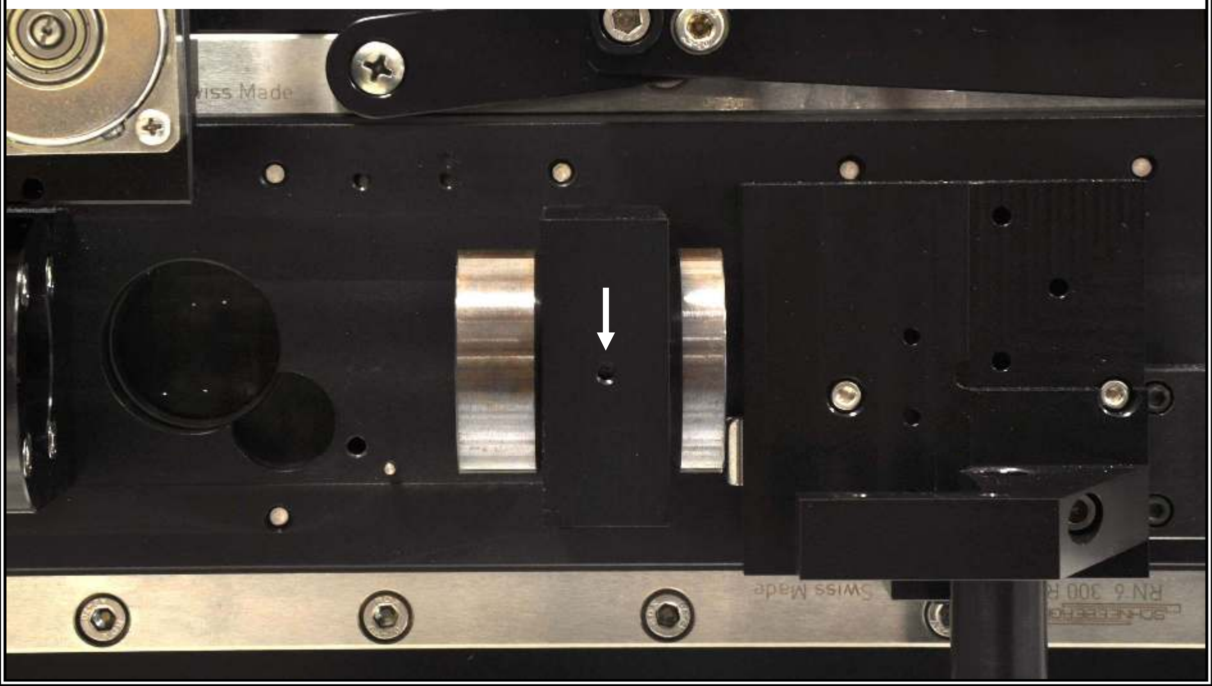


The correctly attached detector-path mirror is shown.



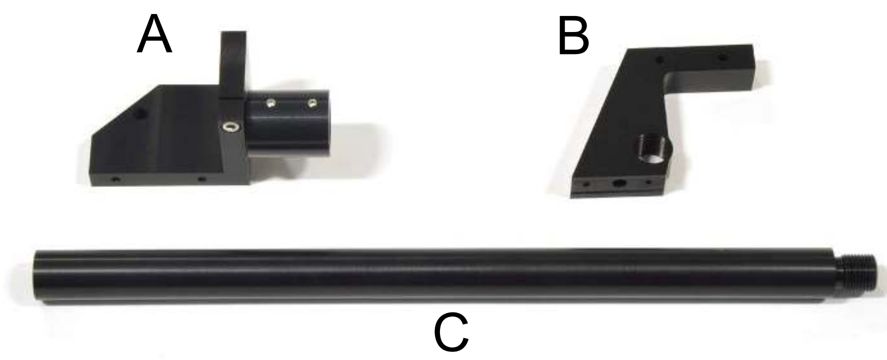
The Leica scan lens is shown. The right side points toward the scan mirrors when correctly installed.

Correct placement of the scan lens (note orientation) is shown.
A set screw in the scan lens mount (arrow) fixes the location.



For resonant scanner mounting instructions, please see Appendix C instead of pages 35-39.

The parts are shown for the vertical laser pathway.

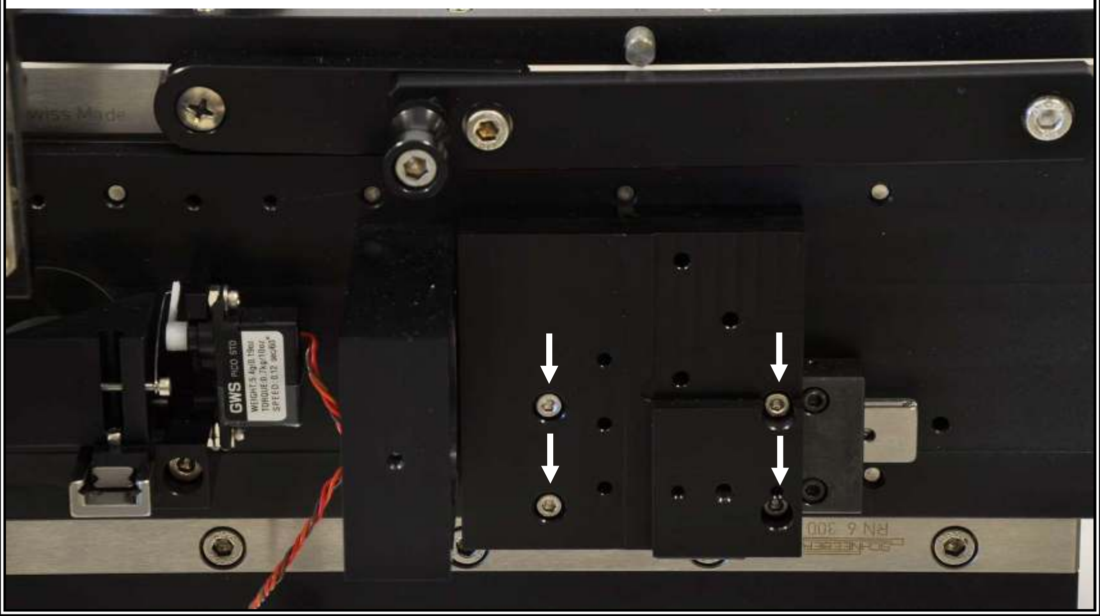


Part A is a bracket for mounting a mirror to turn the laser toward the scanners. Part B is the scanner stimulus bracket. Part C is the laser up tube. Together they form the laser up tube assembly shown below. Part A and B should lie flat at the indicated areas when adjusted properly for rotation.

Attach the NewFocus mirror mount to the bracket at the bottom of the laser entry tube with a single M4X8mm screw. Adjust to a 45 degree angle.

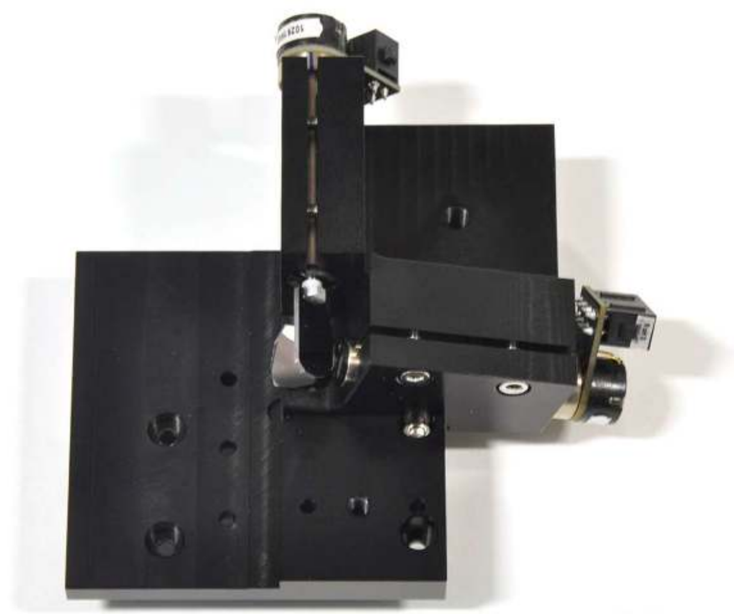
Remove the front surface mirror from its packaging and mount on the NewFocus mirror mount.

Unscrew the scanner block from the side of the Slider by removing the four indicated screws (arrows).

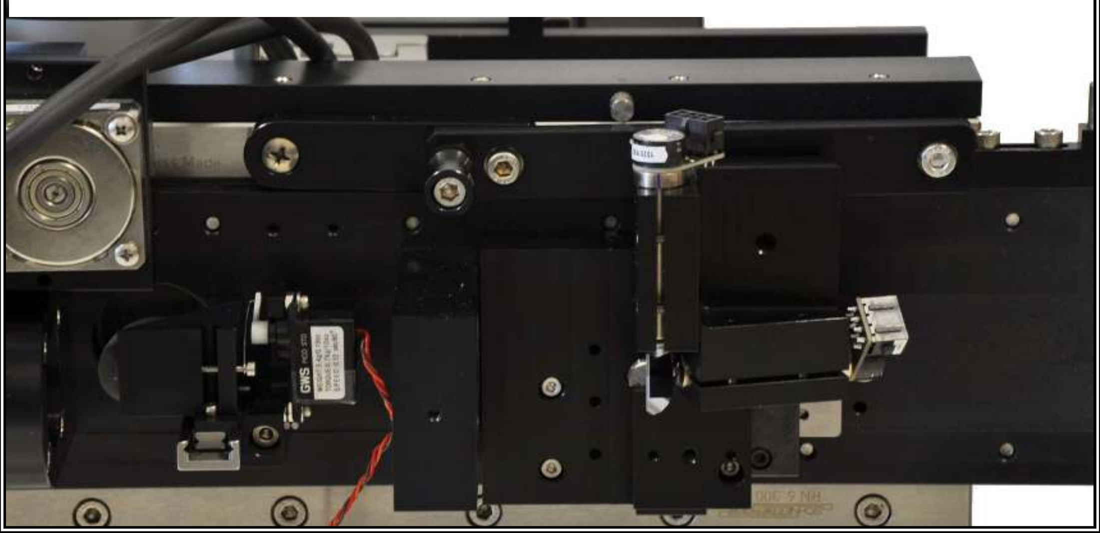


Below is the separated scanner block. It is necessary to leave the indicated (arrow) screw in place while attaching the scan mirrors.

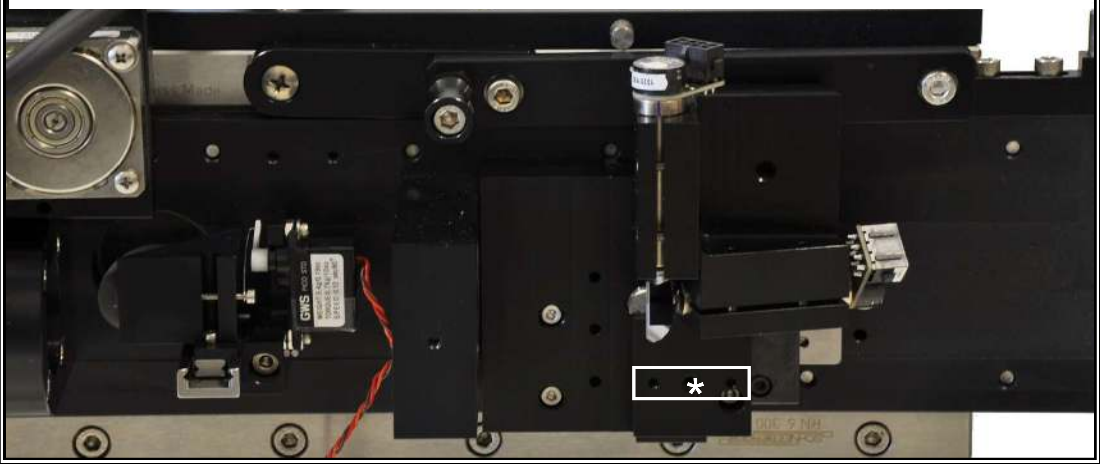
Attach the scan mirrors with two M3x18mm screws and one M3x6mm screw through the back of the scanner block. The small screw is used in the hole beneath the mirrors.



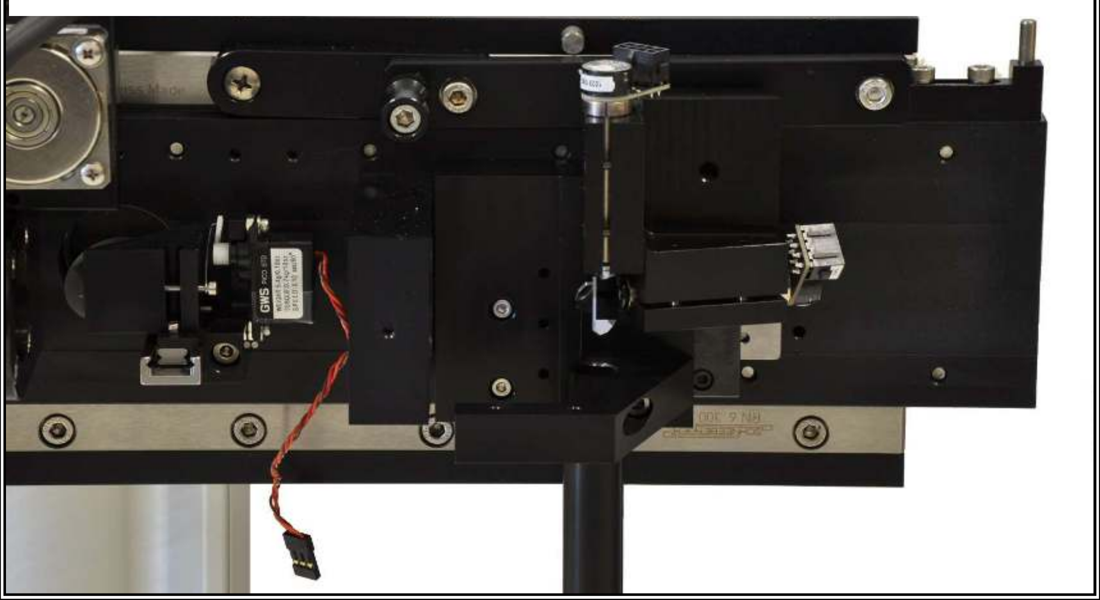
Reattach the scanner block to the Slider with the four M3x6mm screws. The scan mirrors are shipped pre-aligned.



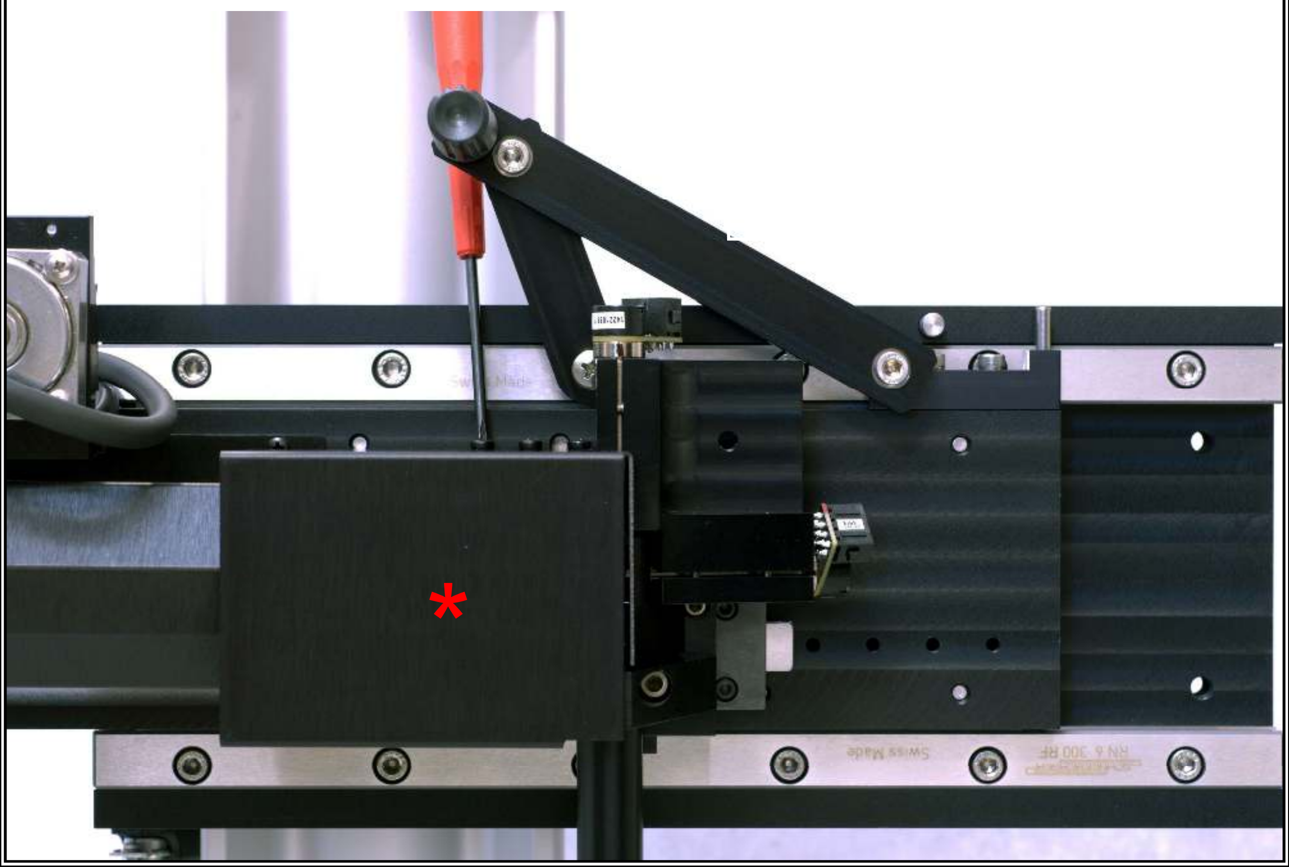
With the scanner block attached, the scanner stimulus bracket and down tube can then be attached. The scanner stimulus bracket is attached with one screw (*) and two alignment pins in the outlined location.



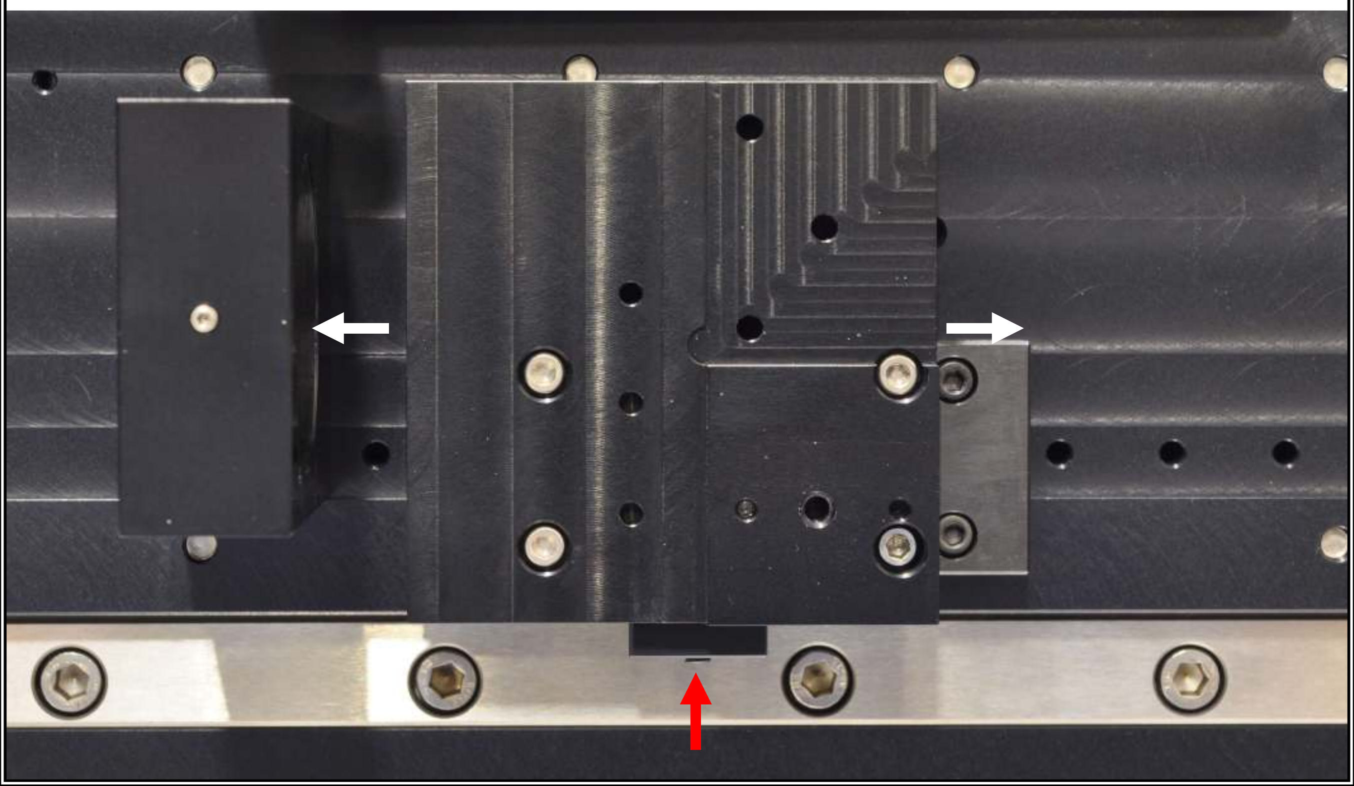
The correctly installed laser up tube pathway is shown. The length of the up tube is dependent on the height of the horizontal laser path and the location of the specimen.



The scanner cover (red astericks) slides from the side of the microscope into the location shown below. The scanner cover is held in place by three screws from above. Access to the screws requires lifting the lever that slides the MOM front to back. The method for accessing the screws is illustrated in the picture below with the red handled screwdriver.



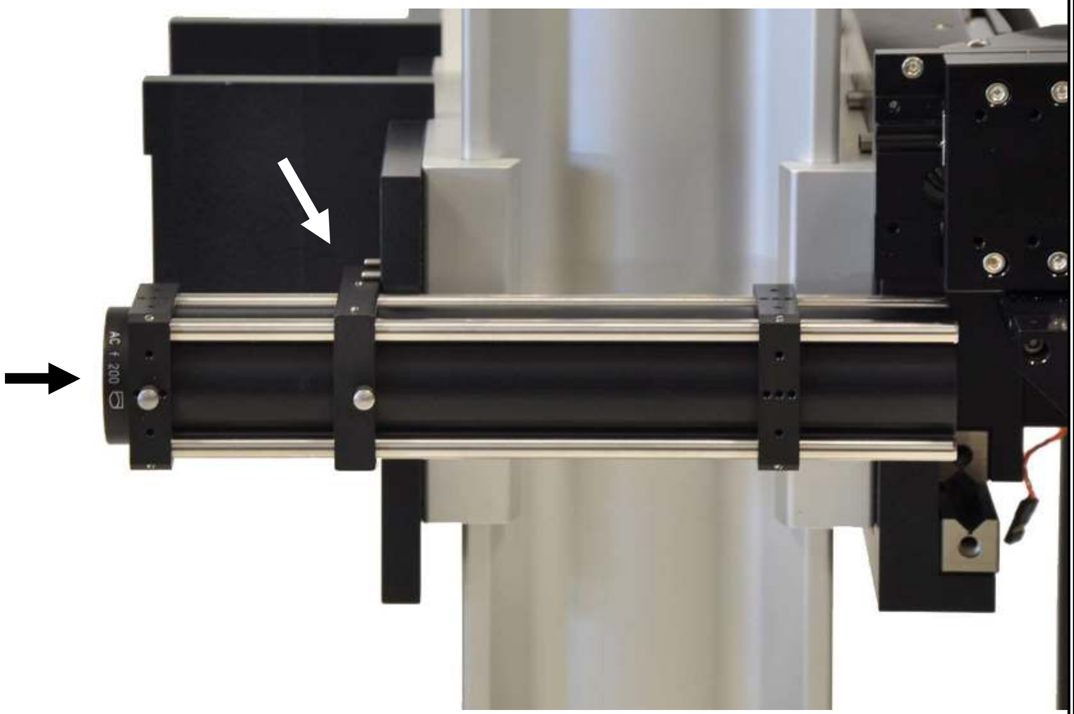
The distance between the scan mirrors and the scan lens is adjusted by sliding the scanner block on its bearing after releasing tension on a set screw (red arrow).



The optical path from the main body of the microscope to the vertical illuminator is shown with the attachment hardware in a bag.

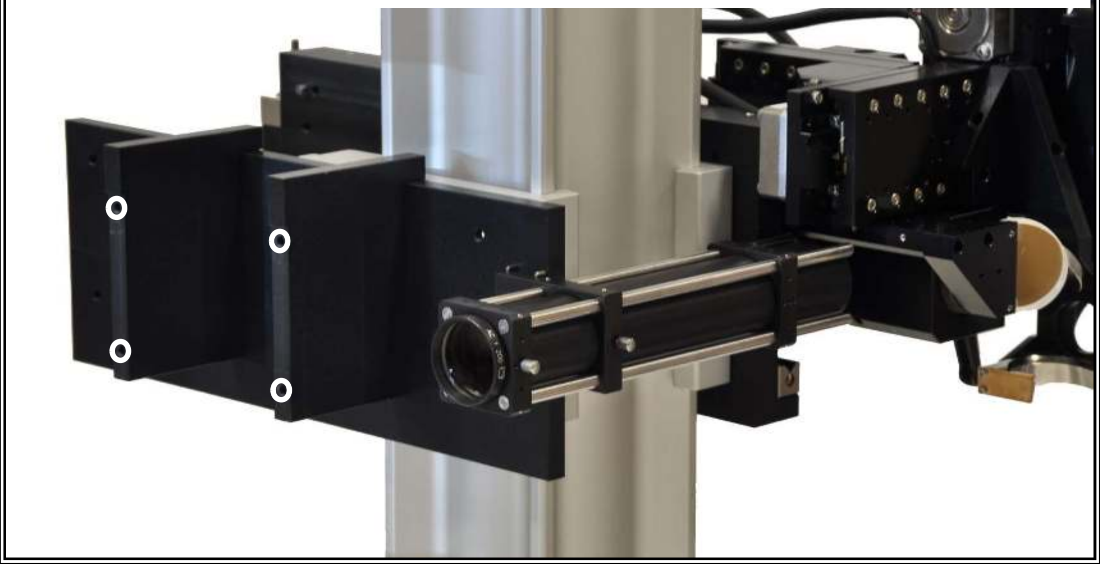
Care must be taken to not touch the attached lens (arrow).

The correct attachment of the CCD light path is shown (front view).

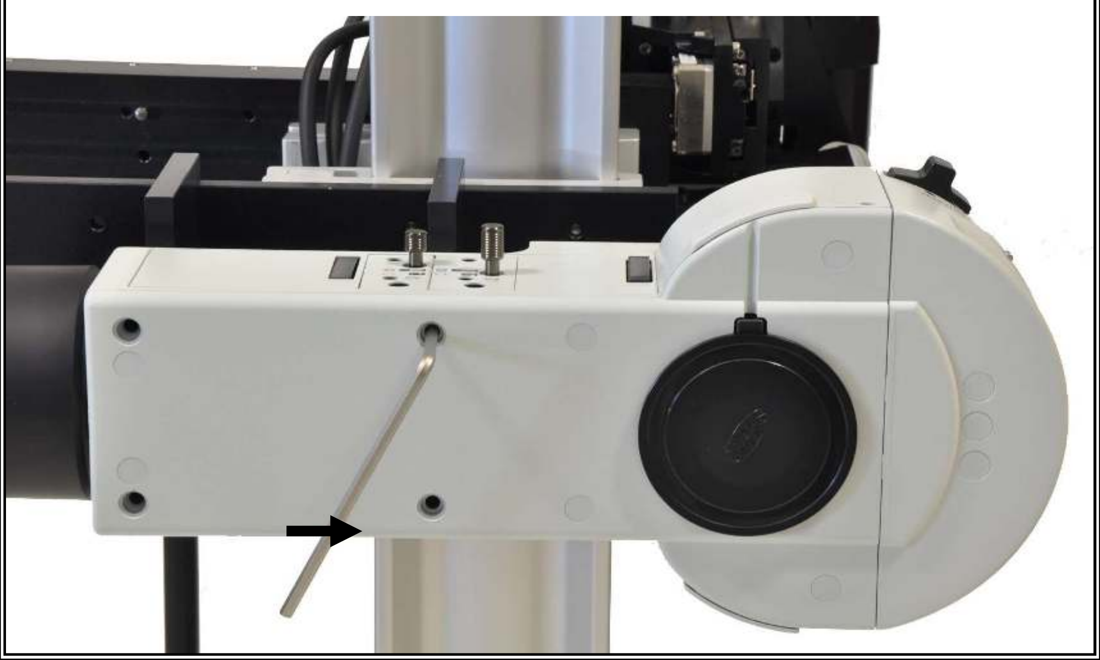


The optical path is attached to the Olympus mount by a single screw (M6X14mm) and is aligned by two pins (white arrow indicates mounting area). With the pathway in place you can align the halves of the microscope to each other by looking through the tube in the direction of the black arrow in the picture. Make both halves the same height by moving the Olympus mount up or down on the X-95 rail. Adjust for concentricity of the pathway.

Attach the Olympus vertical illuminator to the Olympus mount by four captive screws inside the illuminator in the outlined areas.



These are turned by a special hex key (arrow) included with the illuminator (side view of microscope).



Correct attachment of the Olympus vertical illuminator to the CCD light path is shown (corner view).

Remove the cover over the attachment location for the Olympus tube lens using a 3mm allen wrench.

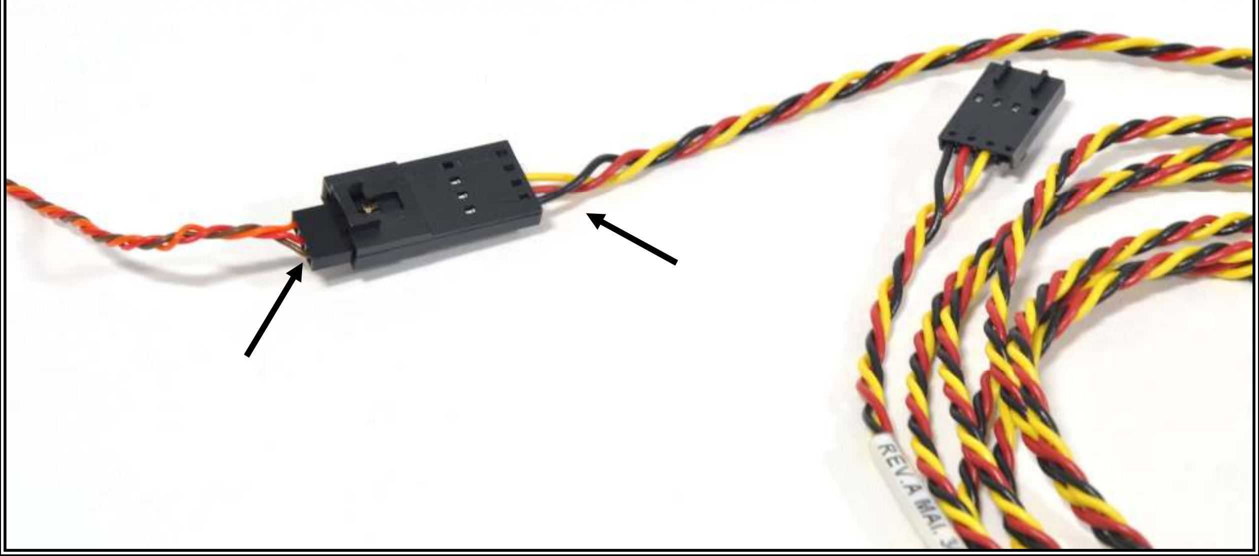
Install the Olympus tube lens into the Olympus vertical illuminator.

Install the Sutter emission adapter with one C-mount extension adapter.

If a trinoc is ordered, the mirror to turn the light path must be installed in the articulating mount shown below.

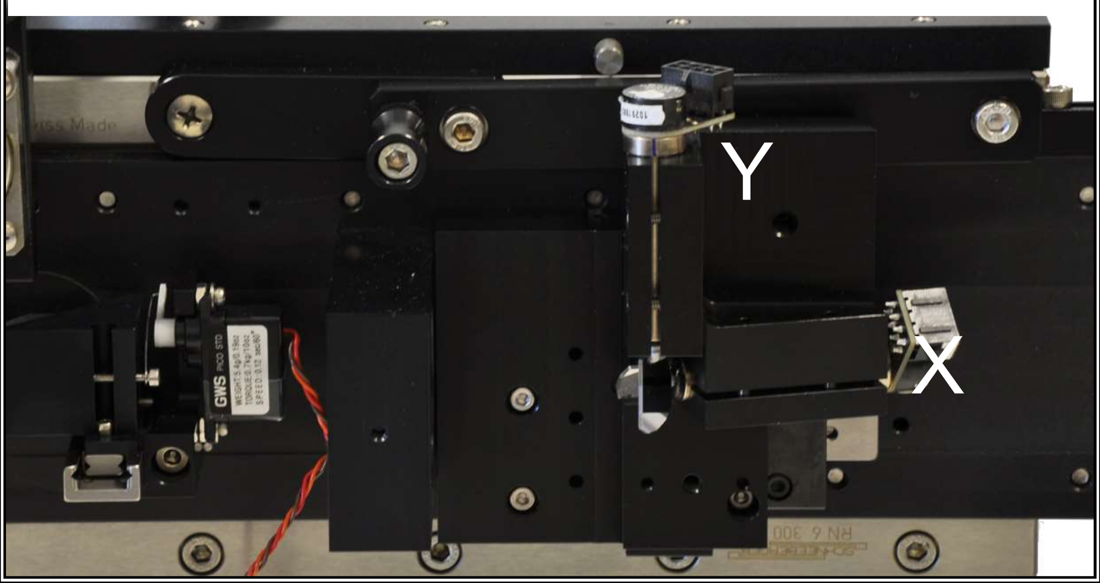
The trinoc is shown correctly installed.

The servo connectors are wired to extension cables. The brown (or black) wire from the servo connects to the black wire in the extension cable as shown below (arrows). The other end of the extension cables plug into the servo control box.



The MDR-6 scan mirror controller box is shipped with 6215 galvanometers. Cables are provided to attach the GALVO X and GALVO Y outputs to the galvonometers.

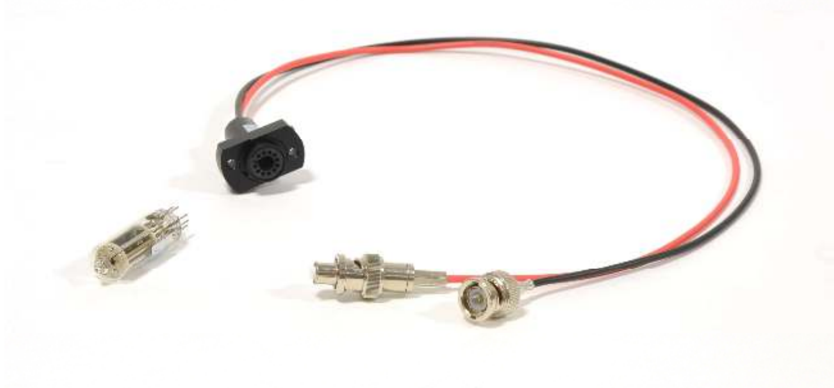
The GALVO X and Y cables should be attached to the appropriate galvonometer as indicated below.

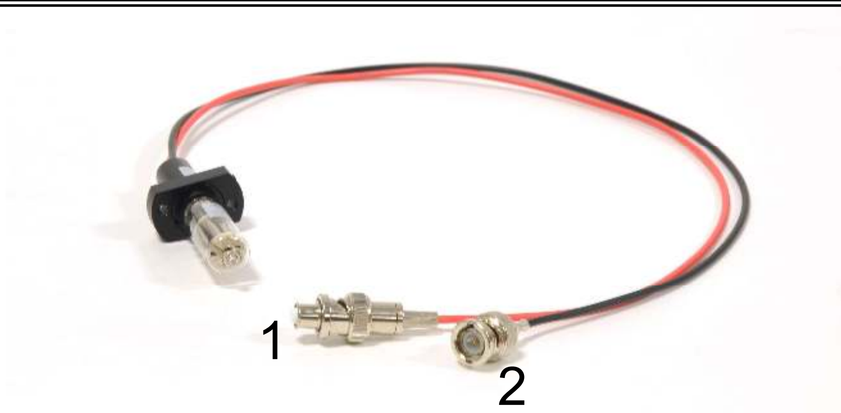


The PS-2 is shipped with side-on PMTs.

The 12 volt output of the PS-2 attaches to the back of the PMT preamplifier shown below.

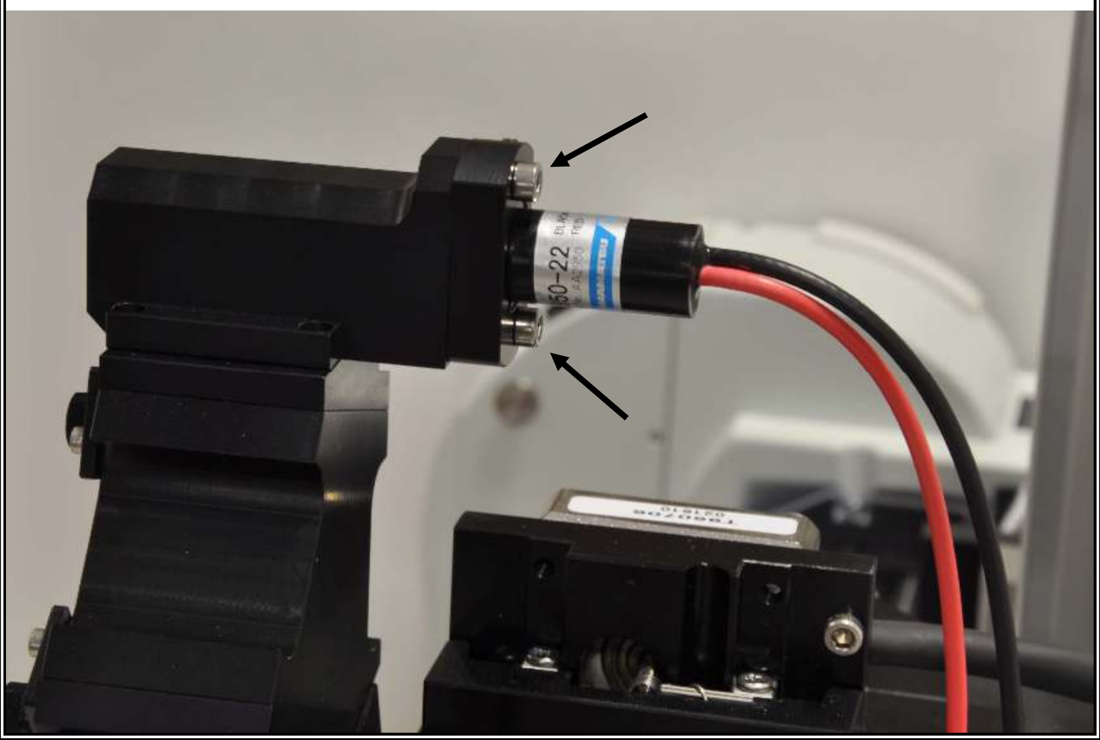
A side-on Hamamatsu PMT is shown below. The PMT must first be connected to the socket.





Connector 1 from above attaches to the HV output of the PS-2 with an extension cable. Connector 2 from above attaches to the front (INPUT) of the PMT preamplifier shown below with an extension cable.

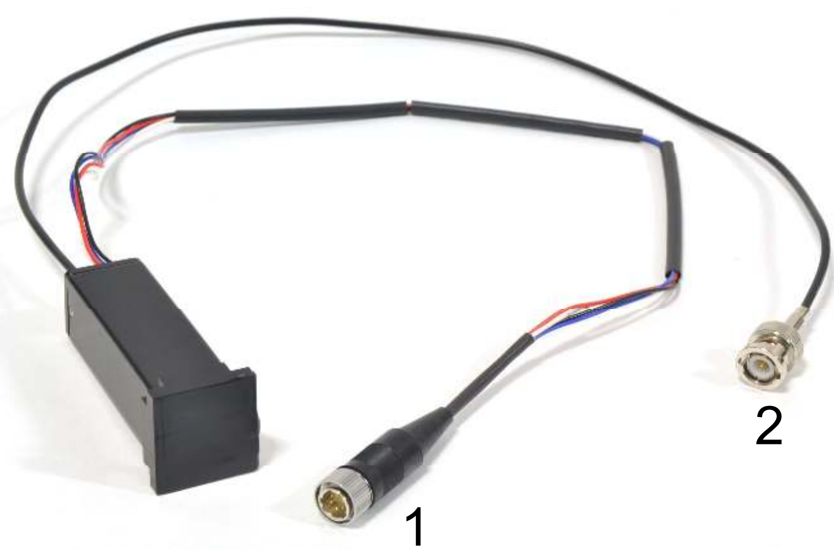
The mount for side-on PMTs is shipped attached to the detection pathway. The PMT in its collar is attached to the mount using two M3x10mm screws (arrows).



The PS-2LV is shipped with end-on GaAsP PMTs.

The 12 volt output of the PS-2LV attaches to the back of the PMT preamplifier shown below.

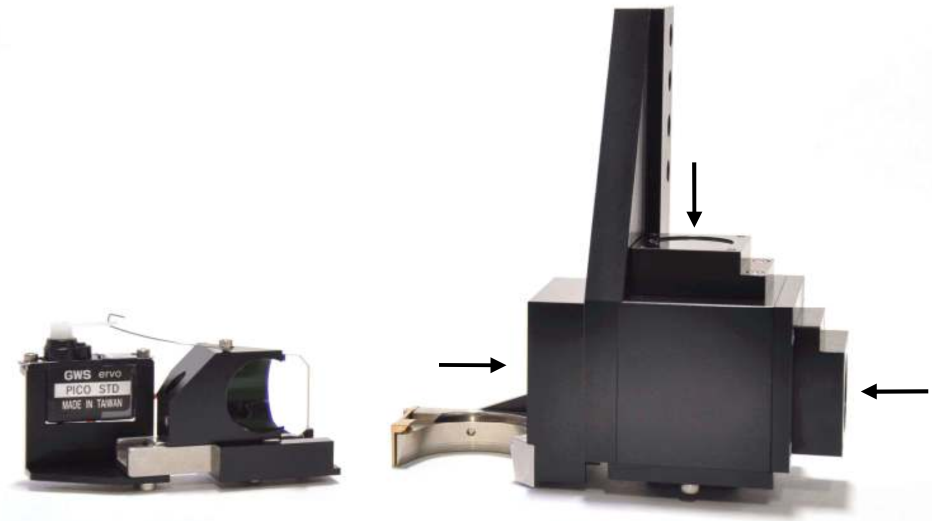
An end-on Hamamatsu GaAsP PMT is shown below. Remove the tape covering the photosensitive portion of the PMT before mounting to the detector pathway with four M2x4mm screws. Care should be taken to minimize light exposure even when not powered.



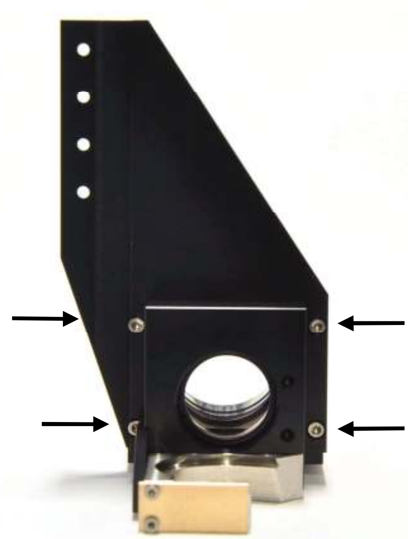
Connector 1 from above attaches to the LV output of the PS-2LV with an extension cable. Connector 2 from above attaches to the front (INPUT) of the PMT preamplifier shown below with an extension cable.

Completed Microscope

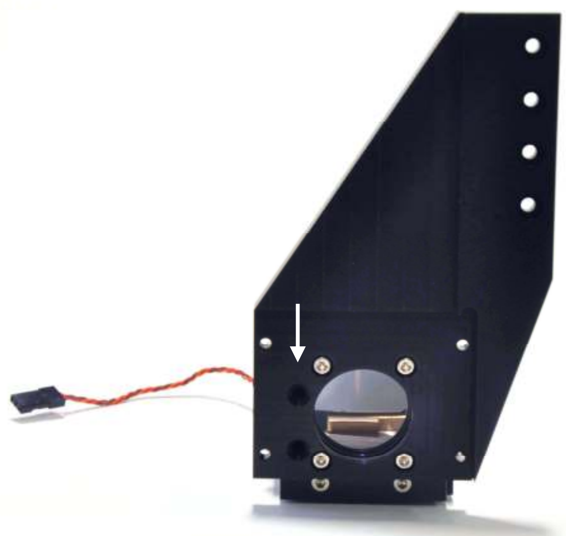
Shown below is the dichroic slider (left) and the optional Short-Path detection assembly (right). The Short-Path detection assembly is shipped with all optics in place. Therefore, it is important to handle it carefully so as not to touch the aspherical collection lenses or the laser blocking filter (arrows).



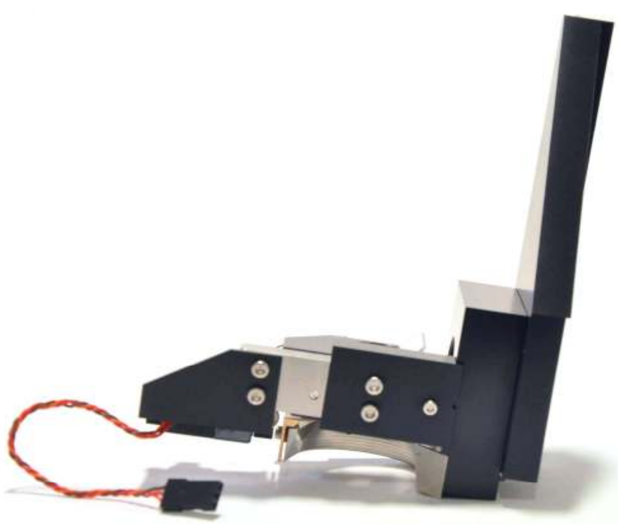
In order to attach the dichroic slider to the Short-Path detection assembly, you must first separate the two halves of the short path by removing four screws (arrows).



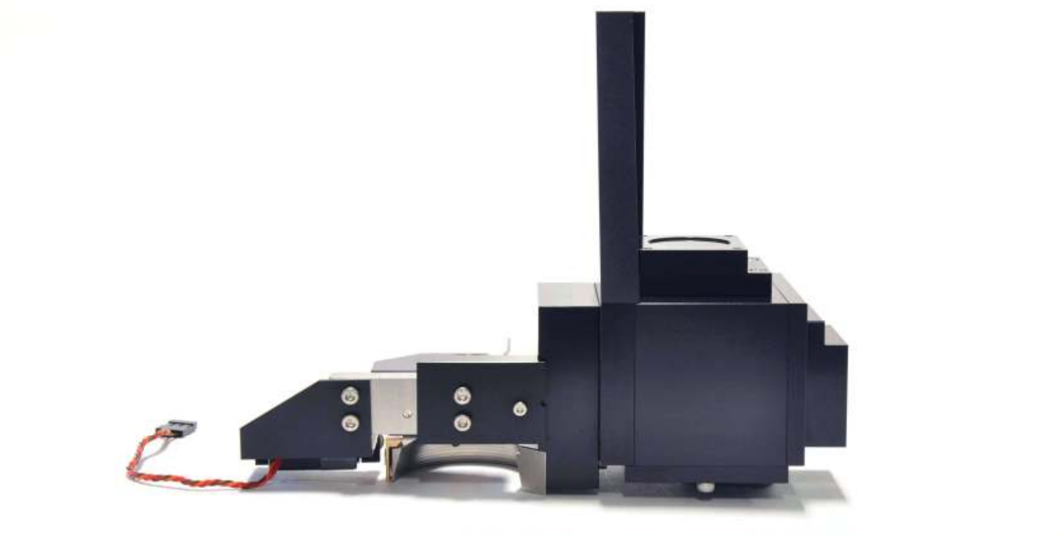
Two captive M3 socket head cap screws are used to attach the dichroic slider to the short path detection assembly. Use a 2.5mm allen wrench to connect the two assemblies. Access is gained to the screws through the two holes shown below (white arrow).



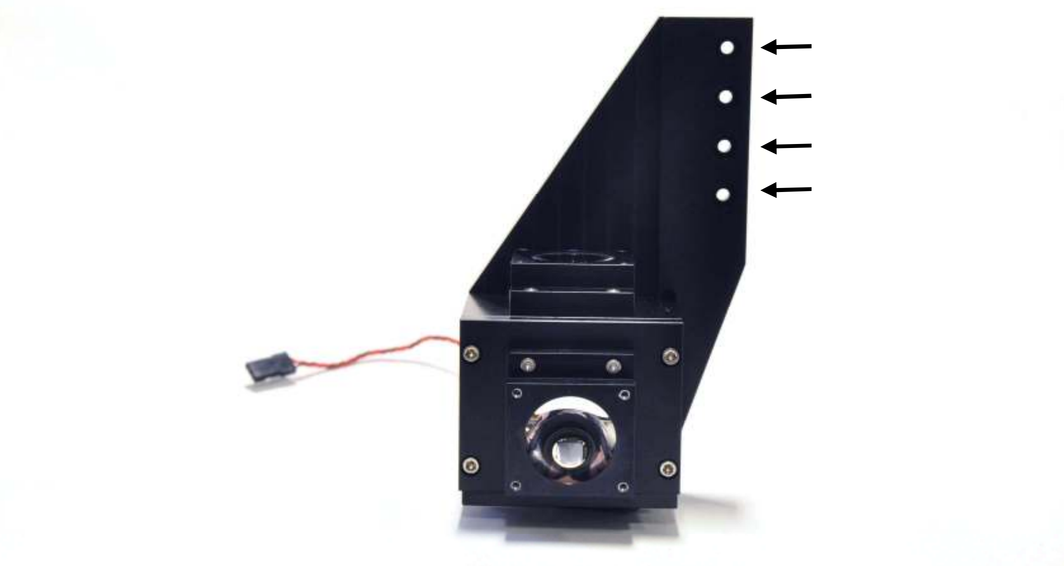
The dichroic slider is shown attached below.



The rest of the Short-Path detection assembly is then reattached with the four screws you removed at the beginning.

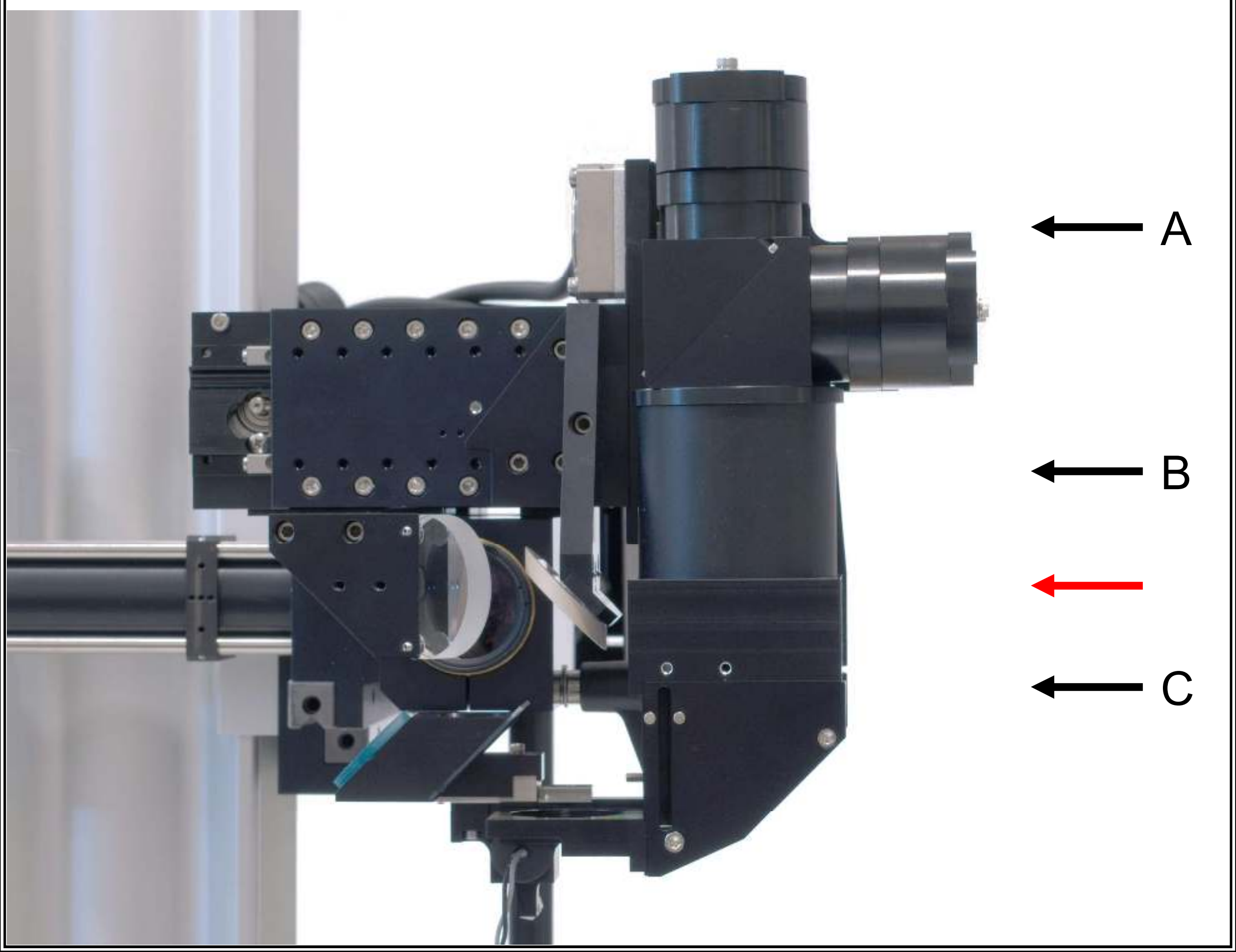


The entire assembly is then attached to the Z axis moving slider with four M3x8mm screws (arrows).

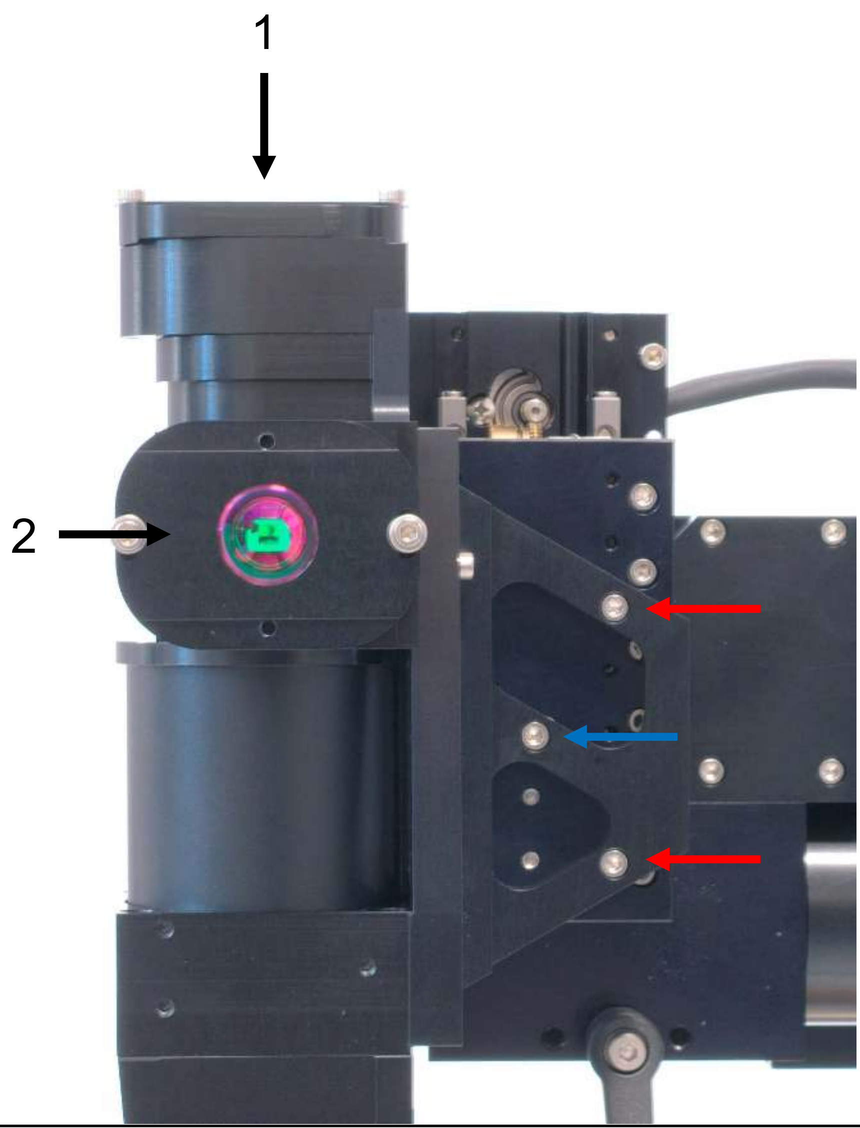


Now return to page

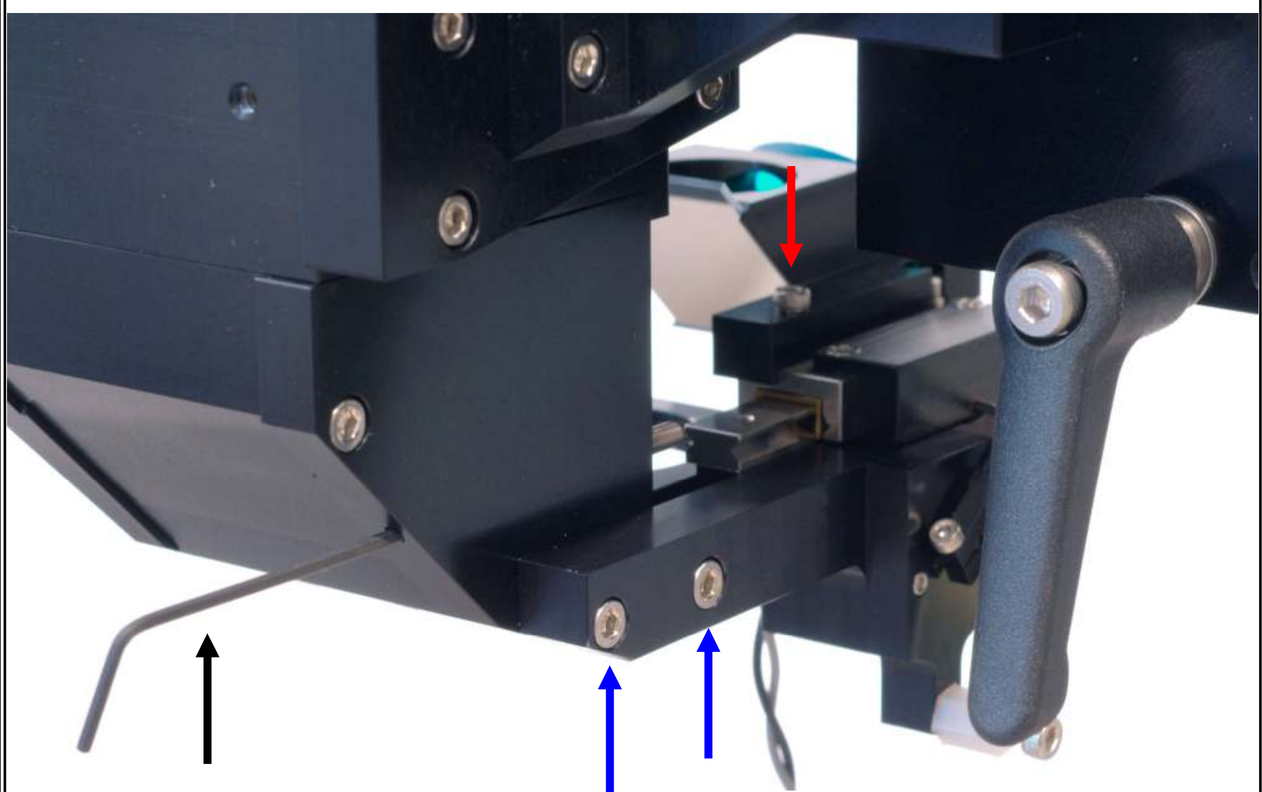
The Wide-Path Detector is shown below correctly mounted. Before attaching the detector head (A), the light baffle (B) and the objective mount/turning mirror (C), the included 2" lens must be placed in position (red arrow) with the belly of the lens facing downward.



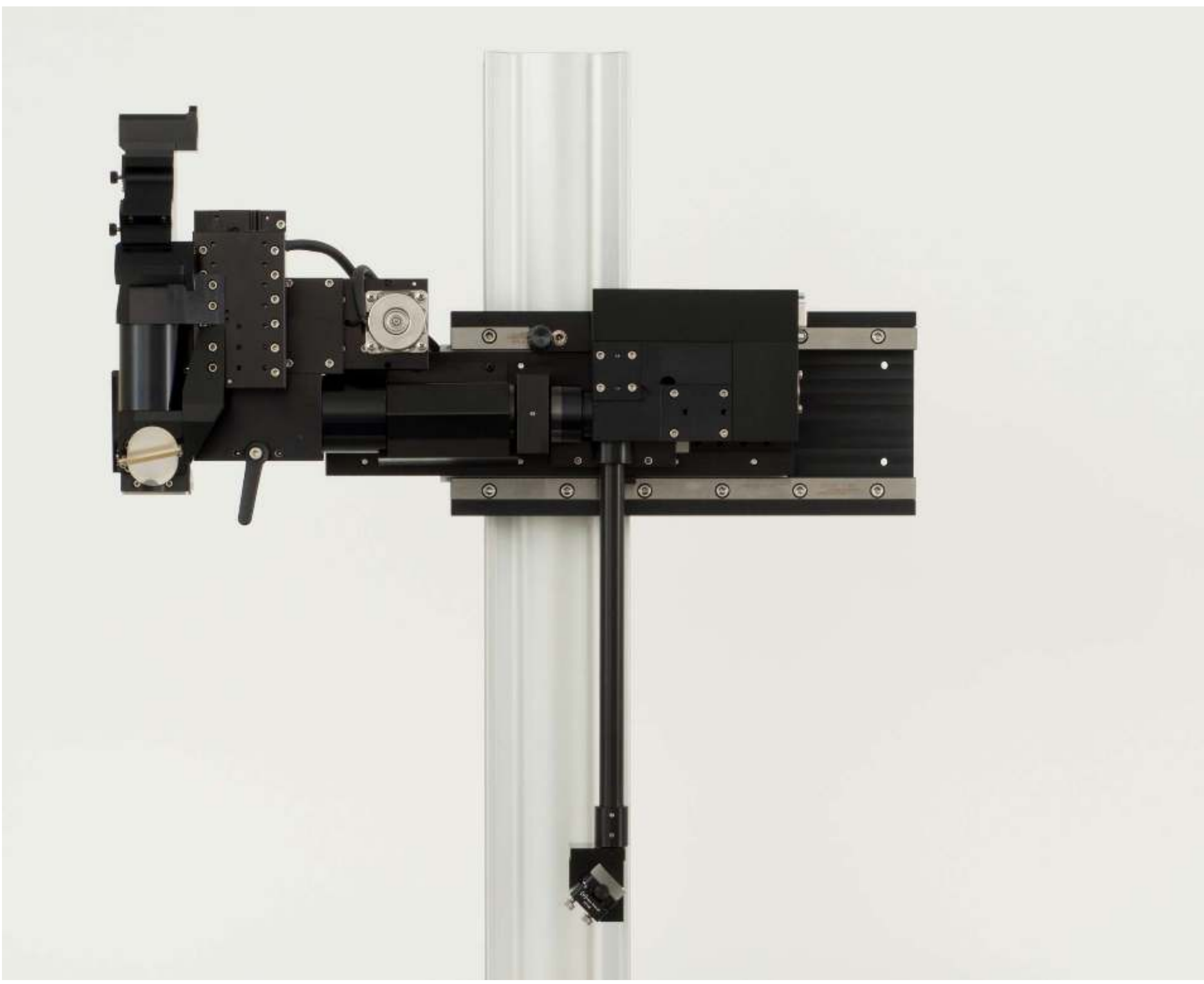
The Wide-Path detector assembly is attached to the Z-slide with two M3x10mm screws (red arrows) and one M3x8mm screw (blue arrow). The light baffle has a cut out section that must be aligned with the edge of the Z-slide. The PMTs are mounted at locations 1 and 2.



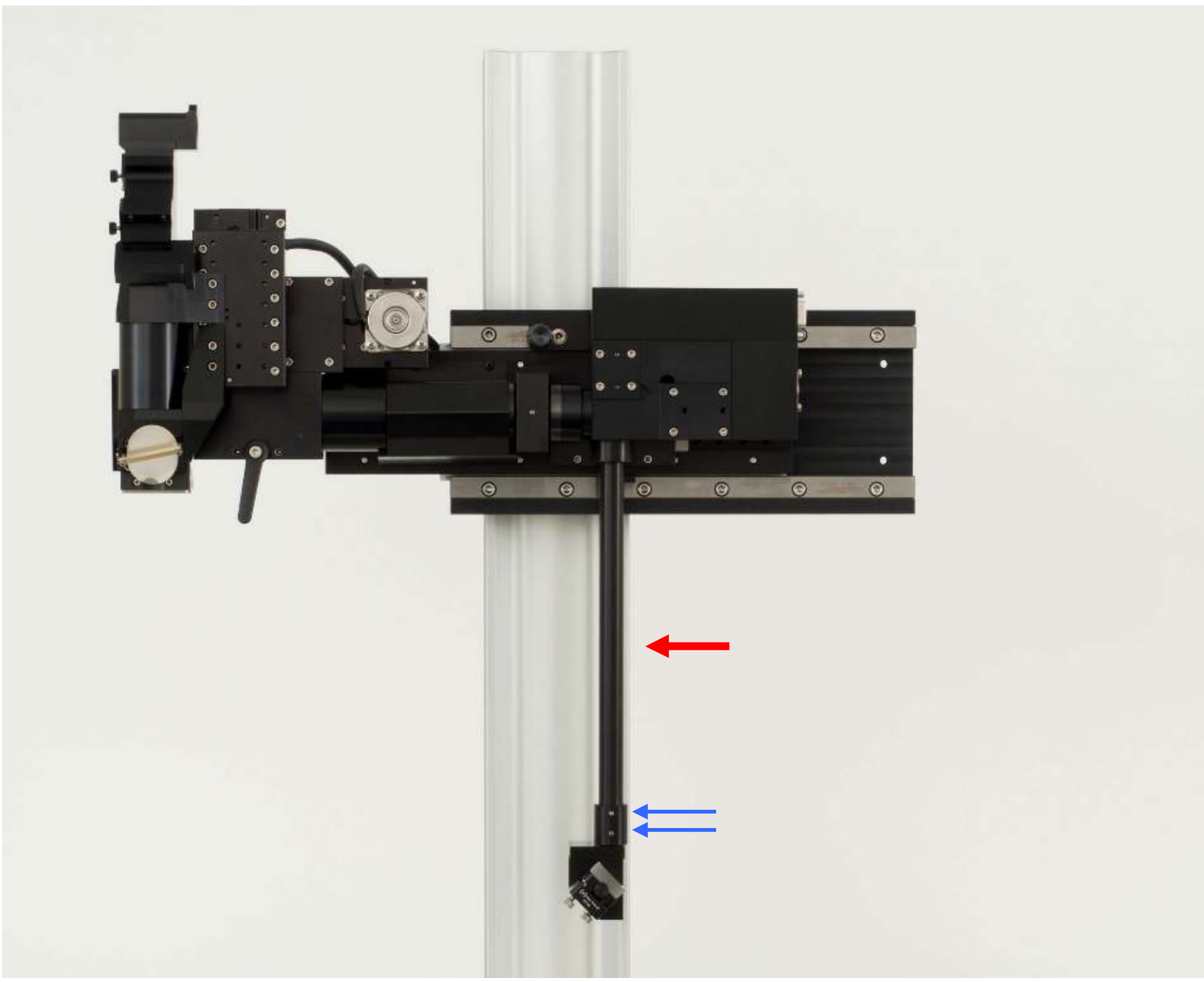
The dichroic is attached to the actuator rail with a single screw as indicated by the red arrow. The complete dichroic actuator assembly is then attached with two screws as indicated with the blue arrows. Attachment of the dichroic actuator is most easily done with the detector on the microscope but the Y and YZ mirrors not yet attached. It is also useful to use the MP285/MPC200 controller to move the X, Y and Z slides to their most expanded positions for easy access. The position of the dichroic mirror over the back aperture of your objective is set by advancing a set screw using an allen wrench as indicated (black arrow). Care should be taken to never move the dichroic by hand as this may damage the actuator mechanism.



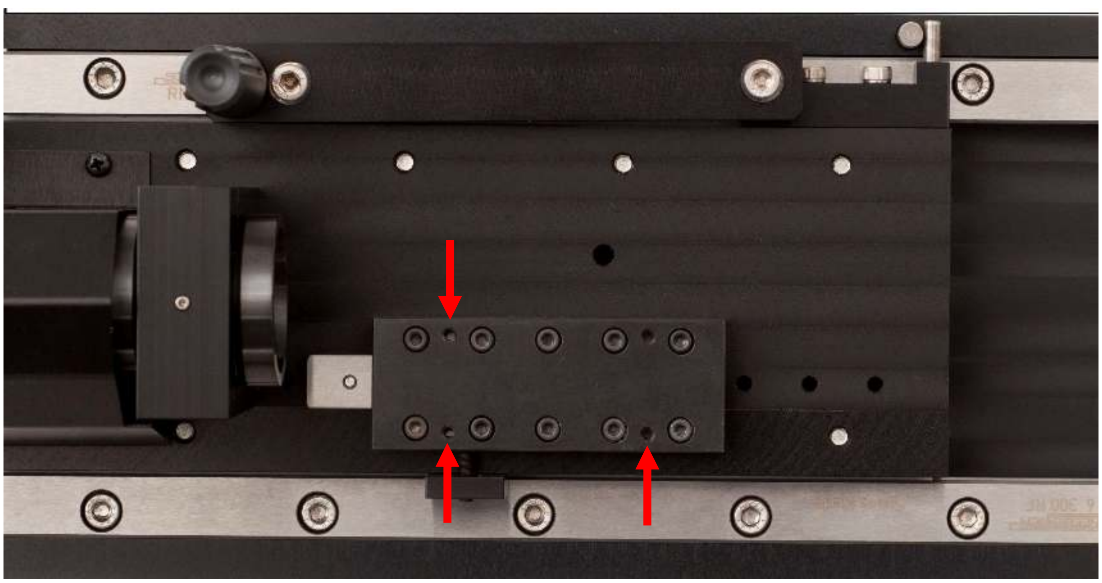
A correctly mounted resonant scanner box is shown below.
Subsequent pages will describe the procedure for
this portion of the assembly.



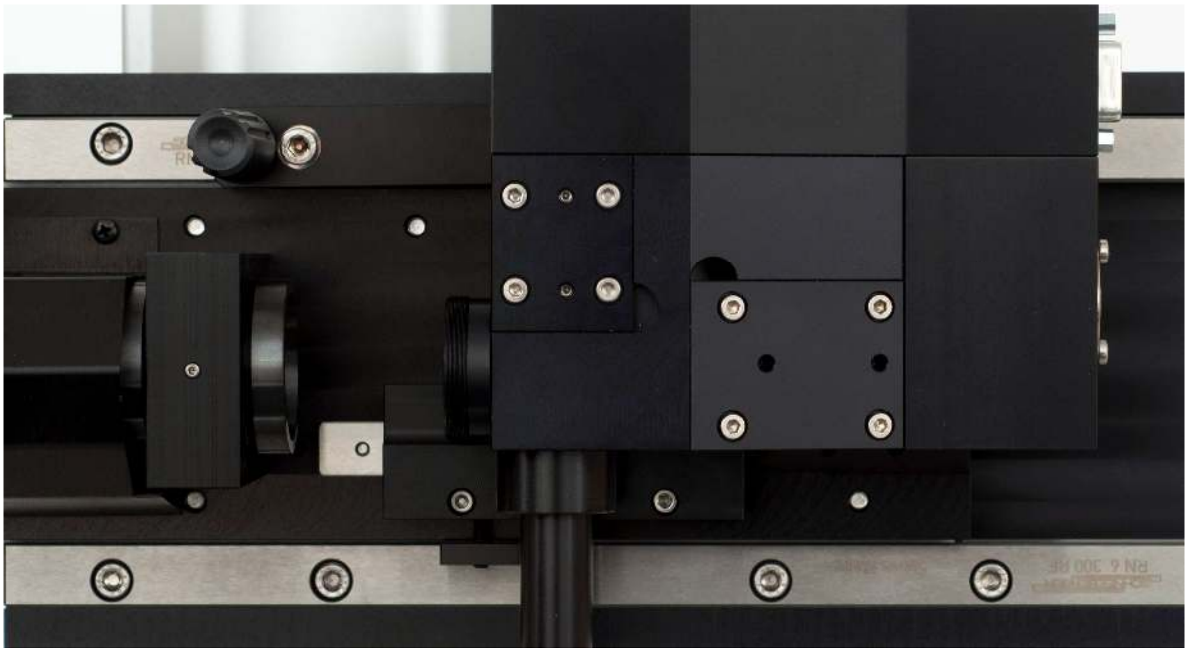
The periscope (red arrow) should be attached to the scanner box prior to mounting the scanner box on the MOM. It is important that the mirror mount at the base of the periscope be oriented such that the incoming beam is parallel to the optical axis of the scan and tube lens in the microscope. The rotation of the mirror mount is set using the two set screws indicated by the blue arrows.



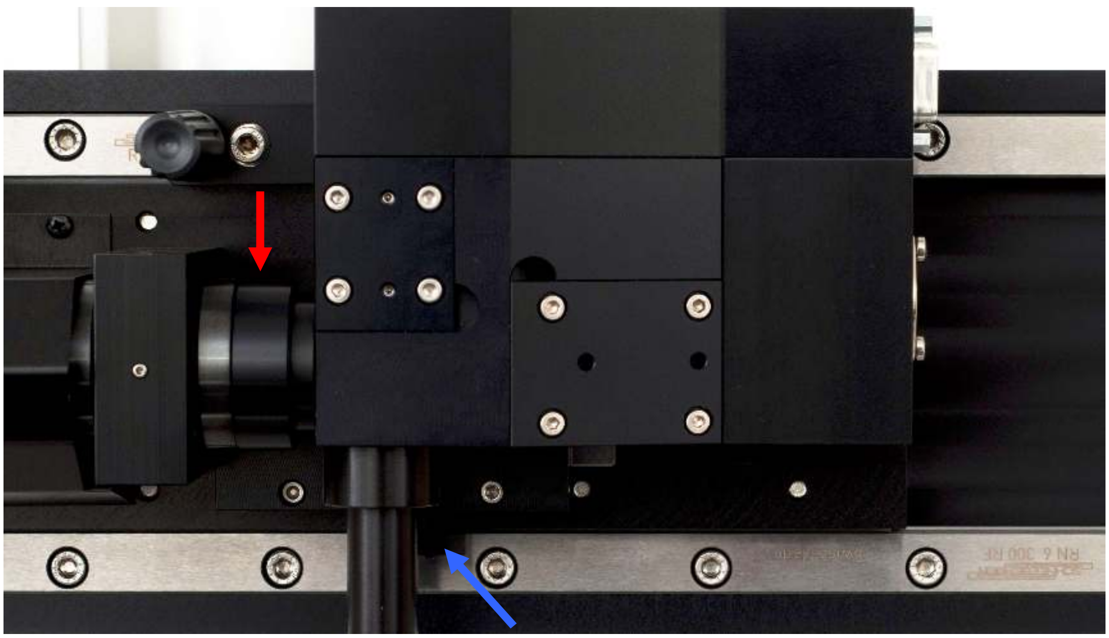
The resonant scaner box is attached to the bearing rail shown below using three M3x6mm screws at the locations indicated with red arrows.



The correct attachment of the resonant scaner box to the bearing rail is illustrated below.



The internally threaded ring is then added to the front of the scan box (red arrow). Once the correct optical position of the scanner box along the travel of the bearing rail is determined, the bearing should be locked in place with the set screw in the location indicated with the blue arrow.



The internally threaded ring should then be rotated until a seal is formed between it and the back of the scan lens. This is important for noise suppression of the resonant scanner.

The terminals at the rear of the resonant scanner box connect to the MDR-R controller with the included cables.